Supplemental Material for "When does tail-risk optimization change a quantum gate? An equivalence boundary for mean and CVaR control"

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Equations, figures, tables and sections of the main text are cited without the prefix S (e.g. Eq. (1), Theorem 1); those of this Supplement carry it. In the code and data files, experiments E1–E6 are called Stage 0, E7–E8 Stage 1 and E9 Stage 2.

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S1. PROOFS

Throughout, Z, Z_1, Z_2 denote independent standard normal variables, φ and Φ the standard normal density and distribution function, and $h(\phi) = \sin^2(\phi/2)$. We use four standard properties of CVaR_α, all immediate from the Rockafellar–Uryasev form (1): *monotonicity* ($X \le Y$ almost surely implies $\mathrm{CVaR}_\alpha X \le \mathrm{CVaR}_\alpha Y$), *translation invariance* ($\mathrm{CVaR}_\alpha(X + c) = \mathrm{CVaR}_\alpha X + c$), *positive homogeneity* ($\mathrm{CVaR}_\alpha(sX) = s\,\mathrm{CVaR}_\alpha X$ for $s \ge 0$) and *subadditivity* ($\mathrm{CVaR}_\alpha(X + Y) \le \mathrm{CVaR}_\alpha X + \mathrm{CVaR}_\alpha Y$) [1, 2]. Subadditivity follows by choosing $\tau = \tau_X + \tau_Y$ in (1) and using $(x+y-\tau_X-\tau_Y)_+ \le (x-\tau_X)_+ + (y-\tau_Y)_+$. CVaR is also *law invariant*: it depends on X only through its distribution.

A. The equivalence theorem

Periodicity. The infidelity h is strictly increasing in $|\phi|$ on $[0, \pi]$, strictly decreasing on $[\pi, 2\pi]$ and 2π-periodic: $h'(\phi) = \frac{1}{2}\sin\phi$, positive on $(0, \pi)$ and negative on $(\pi, 2\pi)$; h is even and 2π-periodic. □

Mean infidelity. $h(\phi) = \frac{1}{2}(1 - \cos\phi)$ and, for $\phi \sim \mathcal{N}(0, \sigma^2)$, $\mathbb{E}\cos\phi = \mathrm{Re}\,\mathbb{E}e^{i\phi} = e^{-\sigma^2/2}$ (the characteristic function of a Gaussian). Hence $\mathbb{E}h(\phi) = \frac{1}{2}(1 - e^{-\sigma^2/2})$, whose derivative in σ^2 is $\frac{1}{4}e^{-\sigma^2/2} > 0$. □

Majorant. $\widetilde{L} \ge h$ everywhere, with equality for $|\phi| \le \pi$, and $\widetilde{L}$ is non-decreasing in $|\phi|$. For $|\phi| \le \pi$, $\widetilde{L} = h$. For $|\phi| > \pi$, $\widetilde{L} = 1 \ge h$. Monotonicity: on $[0, \pi]$, $\widetilde{L} = h$ increases from 0 to 1 (periodicity, above); on (π, ∞) it is constant at its maximum value 1. □

Sandwich bound, Eq. (3). Let $E = \{|\phi| > \pi\}$ and $p = \Pr(E)$. *Upper bound*: $L \le \widetilde{L}$ pointwise , so $\mathrm{CVaR}_\alpha L \le \mathrm{CVaR}_\alpha \widetilde{L}$ by monotonicity. *Lower bound*: pointwise, $\widetilde{L} \le L + \mathbf{1}_E$, since on E^c the two sides are equal up to the nonnegative indicator, and on E the left side is 1 while the right side is at least 1. Monotonicity and subadditivity give $\mathrm{CVaR}_\alpha \widetilde{L} \le \mathrm{CVaR}_\alpha L + \mathrm{CVaR}_\alpha \mathbf{1}_E$. For the Bernoulli variable $\mathbf{1}_E$, Eq. (1) gives

$$\mathrm{CVaR}_\alpha \mathbf{1}_E = \min_\tau \Big\{ \tau + \tfrac{1}{\alpha}\big[p(1-\tau)_+ + (1-p)(-\tau)_+\big] \Big\} = \min\Big(1, \frac{p}{\alpha}\Big) \le \frac{p}{\alpha},$$

where the minimum is attained at $\tau = 0$ if $p \le \alpha$ and at $\tau = 1$ otherwise (the objective is piecewise linear in τ with kinks at 0 and 1). Rearranging, $\mathrm{CVaR}_\alpha L \ge \mathrm{CVaR}_\alpha \widetilde{L} - p/\alpha$. No condition on p versus α is needed, because the bound $\min(1, p/\alpha) \le p/\alpha$ holds always. □

Theorem 1. Write $\phi_i = \sigma_{f_i} Z$ (same Z; allowed because CVaR is law invariant). By the lower half of the sandwich bound for f_2 and the upper half for f_1,

$$\mathrm{CVaR}_\alpha L(\phi_2) - \mathrm{CVaR}_\alpha L(\phi_1) \ge \big[\mathrm{CVaR}_\alpha \widetilde{L}(\phi_2) - r_\alpha(\sigma_{f_2})\big] - \mathrm{CVaR}_\alpha \widetilde{L}(\phi_1) = \Delta_{\mathrm{mono}} - r_\alpha(\sigma_{f_2}).$$

$\Delta_{\mathrm{mono}} \ge 0$: since $\sigma_{f_2} > \sigma_{f_1}$, $|\phi_2| \ge |\phi_1|$ for every realization of Z, and $\widetilde{L}$ is non-decreasing in $|\phi|$, so $\widetilde{L}(\phi_2) \ge \widetilde{L}(\phi_1)$ pointwise; apply monotonicity. For the residual, $\Pr(|\phi| > \pi) = 2\Pr(Z > \pi/\sigma)$ and the Mills-ratio bound $\Pr(Z > x) \le \varphi(x)/x$ for $x > 0$ (which follows from $\int_x^\infty \varphi(t)\,dt \le \int_x^\infty \frac{t}{x}\varphi(t)\,dt = \varphi(x)/x$) gives

$$r_\alpha(\sigma) = \frac{2\Pr(Z > \pi/\sigma)}{\alpha} \le \frac{2}{\alpha}\,\frac{\sigma}{\pi}\,\frac{e^{-\pi^2/(2\sigma^2)}}{\sqrt{2\pi}} = \frac{1}{\alpha}\sqrt{\frac{2}{\pi}}\,\frac{\sigma}{\pi}\,e^{-\pi^2/(2\sigma^2)}. \qquad \square$$

When is $\Delta_{\mathrm{mono}} > 0$ strictly? If $\sigma_{f_1} < \sigma_{f_2}$ and $\Pr(|\phi_2| < \pi) > 0$ (always true for a Gaussian), then $\widetilde{L}(\phi_2) > \widetilde{L}(\phi_1)$ on the event $\{0 < |\phi_2| < \pi\}$, which has positive probability and includes part of the upper α tail of $\widetilde{L}(\phi_2)$ whenever $\Pr(|\phi_2| > \pi) < \alpha$. The CVaR difference is then strictly positive. In the small-noise regime it equals $c_\alpha(\sigma_{f_2}^2 - \sigma_{f_1}^2) + O(\sigma^4)$ by Proposition 1.

Optimizer equivalence. Claim: let $u^\star$ minimize $\mathbb{E}L(\phi[u])$. If $\Delta_{\mathrm{mono}}(\sigma_{u^\star}, \sigma_u) > r_\alpha(\sigma_u)$ for every control with $\sigma_u > \sigma_{u^\star}$, then $u^\star$ is a global CVaR minimizer, and every CVaR minimizer has the same loss distribution. *Proof:* By the mean formula, $\mathbb{E}L(\phi[u])$ is a strictly increasing function of σ_u, so $u^\star$ minimizes σ_u. Take any control u. If $\sigma_u > \sigma_{u^\star}$, Theorem 1 and the hypothesis give $\mathrm{CVaR}_\alpha L(u) - \mathrm{CVaR}_\alpha L(u^\star) \ge \Delta_{\mathrm{mono}} - r_\alpha > 0$. If $\sigma_u = \sigma_{u^\star}$, the two losses have the same distribution and the same CVaR. Hence $u^\star$ is a global CVaR minimizer, and every CVaR minimizer has $\sigma_u = \sigma_{u^\star}$. □

Size of the residual. At $\sigma^2 = 0.05$, $\pi/\sigma = 14.05$ and $\pi^2/(2\sigma^2) = 98.7$, so $2\Pr(Z > 14.05) \approx 7.7 \times 10^{-45}$ and $r_{0.05} \approx 1.5 \times 10^{-43}$.

Proposition 1. $h(\phi) = \phi^2/4 - \phi^4/48 + \ldots$, so $|h(\sigma Z) - \sigma^2 Z^2/4| \le \sigma^4 Z^4/48$. CVaR is 1-Lipschitz in the sense $|\mathrm{CVaR}_\alpha X - \mathrm{CVaR}_\alpha Y| \le \mathrm{CVaR}_\alpha |X - Y|$ (monotonicity plus subadditivity), and $\mathrm{CVaR}_\alpha(Z^4) < \infty$, so $\mathrm{CVaR}_\alpha h(\sigma Z) = \frac{\sigma^2}{4}\mathrm{CVaR}_\alpha(Z^2) + O(\sigma^4)$. The upper α tail of Z^2 is the event $\{|Z| > z_\alpha\}$, $z_\alpha = \Phi^{-1}(1 - \alpha/2)$, of probability exactly α, so

$$\mathrm{CVaR}_\alpha(Z^2) = \frac{1}{\alpha}\,2\int_{z_\alpha}^\infty t^2\varphi(t)\,dt = \frac{2}{\alpha}\Big[z_\alpha\varphi(z_\alpha) + \big(1 - \Phi(z_\alpha)\big)\Big] = 1 + \frac{2z_\alpha\varphi(z_\alpha)}{\alpha},$$

since $-t\varphi(t) + \Phi(t)$ is an antiderivative of $t^2\varphi(t)$ (integrate by parts with $\varphi' = -t\varphi$). Numerically, for $\alpha = 0.20, 0.10, 0.05, 0.025, 0.01$: $z_\alpha = 1.282, 1.645, 1.960, 2.241, 2.576$ and $\mathrm{CVaR}_\alpha(Z^2) = 3.25, 4.39, 5.58, 6.80, 8.45$. The mean has $\mathbb{E}Z^2 = 1$, hence the ratios. □

Average gate fidelity. For a unitary error U_{err} on a d-level system, $F_{\mathrm{avg}} = (dF_e + 1)/(d + 1)$ with entanglement fidelity $F_e = |\mathrm{Tr}\,U_{\mathrm{err}}/d|^2$ [3]. For $U_{\mathrm{err}} = e^{-i\phi Z/2}$ and $d = 2$, $F_e = \cos^2(\phi/2)$, so $1 - F_{\mathrm{avg}} = \frac{2}{3}(1 - \cos^2(\phi/2)) = \frac{2}{3}h(\phi)$. Taking expectations with the mean formula gives $\frac{1}{3}(1 - e^{-\sigma_f^2/2})$. Since $\frac{2}{3}h$ is a positive multiple of h, every step of Theorem 1 goes through unchanged. □

Temporal correlation. Claim: for any stationary Gaussian noise, e.g. Ornstein–Uhlenbeck noise $C(t,t') = \sigma^2 e^{-|t-t'|/\tau_c}$, $\phi[u]$ is exactly Gaussian and Theorem 1 applies verbatim. *Proof:* $\phi[f] = \int f\eta\,dt$ is a continuous linear functional of a Gaussian process, hence Gaussian, with variance $\iint f(t)C(t,t')f(t')\,dt\,dt'$. For $f \equiv 1$ and $C = \sigma^2 e^{-|t-t'|/\tau_c}$,

$$\sigma_f^2 = \sigma^2\int_0^{T_g}\int_0^{T_g} e^{-|t-t'|/\tau_c}dt\,dt' = 2\sigma^2\int_0^{T_g}(T_g - s)e^{-s/\tau_c}ds = 2\sigma^2\tau_c\big(T_g - \tau_c(1 - e^{-T_g/\tau_c})\big).$$

As $\tau_c \to \infty$ this tends to $\sigma^2 T_g^2$, the quasi-static limit σ_ϕ^2 used in this paper. □

Several channels on one axis. If $\boldsymbol{\eta}(t)$ is a jointly Gaussian vector process and $\phi = \sum_j \int f_j \eta_j \, dt$, then ϕ is a linear functional of a Gaussian process, hence Gaussian, with variance

$$Q(f) = \sum_{jk} \iint f_j(t) C_{jk}(t,t') f_k(t') \, dt \, dt'. \qquad \square$$

Corollary 1. Let $\tau(u_1) \le \tau(u_2)$ and $L_{u_i} = h(\tau(u_i)Z)$ with h non-decreasing in $|\cdot|$. Then $L_{u_1} \le L_{u_2}$ pointwise on a common probability space, which is equivalent to $L_{u_1} \le_{\mathrm{st}} L_{u_2}$ [4]. Any risk measure that is monotone and law invariant therefore satisfies $\varrho(L_{u_1}) \le \varrho(L_{u_2})$, so all such measures are minimized by the controls with the smallest $\tau(u)$. $\square$

B. Two non-parallel channels

Reduction to two weights. To leading order the error of a draw is $\boldsymbol{\epsilon} = \delta\,\mathbf{b} + \gamma\,\mathbf{a}$ and the loss is $\boldsymbol{\epsilon}^{\mathsf{T}} W \boldsymbol{\epsilon} = \mathbf{x}^{\mathsf{T}} M \mathbf{x}$ with $\mathbf{x} = (\delta/\sigma_\delta, \gamma/\sigma_\gamma)$ and $M = G^{\mathsf{T}} W G$, $G = (\sigma_\delta \mathbf{b}, \sigma_\gamma \mathbf{a})$. With $\mathbf{x} = C^{1/2}\mathbf{y}$, $\mathbf{y}$ standard normal and C the correlation matrix, $L = \mathbf{y}^{\mathsf{T}} C^{1/2} M C^{1/2} \mathbf{y}$. Diagonalizing the symmetric positive semidefinite matrix $C^{1/2} M C^{1/2}$ by an orthogonal matrix (which maps $\mathbf{y}$ to another standard normal vector) gives $L = \lambda_1 Z_1^2 + \lambda_2 Z_2^2$.

Proposition 2(ii), Schur-convexity. Fix τ and write $f_\tau(\lambda) = \tau + \alpha^{-1}\mathbb{E}(\lambda_1 X_1 + \lambda_2 X_2 - \tau)_+$ with $X_i = Z_i^2$. For each outcome, $\lambda \mapsto (\lambda \cdot X - \tau)_+$ is convex (a convex function of an affine function), and expectation preserves convexity. Since X_1, X_2 are i.i.d., $f_\tau(\lambda_1, \lambda_2) = f_\tau(\lambda_2, \lambda_1)$. A symmetric convex function is Schur-convex [5, Ch. 3]: if λ majorizes μ (same sum, μ more equal), then μ is a convex combination of λ and its permutation, $\mu = t\lambda + (1-t)\lambda^\pi$, so $f_\tau(\mu) \le t f_\tau(\lambda) + (1-t) f_\tau(\lambda^\pi) = f_\tau(\lambda)$. This holds for every τ, so taking the minimum over τ on both sides gives $\mathrm{CVaR}_\alpha(L_\mu) \le \mathrm{CVaR}_\alpha(L_\lambda)$. $\square$

The two extreme values. *Equal split*, $\lambda = (\frac{1}{2}, \frac{1}{2})$: $L = \frac{1}{2}(Z_1^2 + Z_2^2)$ is exponential with unit mean. By memorylessness, $\mathbb{E}[L \,|\, L > q] = q + 1$, and $\Pr(L > q) = \alpha$ at $q = \ln(1/\alpha)$, so $\mathrm{CVaR}_\alpha = 1 + \ln(1/\alpha)$, which is 4.00 at $\alpha = 0.05$. *One axis*, $\lambda = (1, 0)$: $L = Z^2$ and, from the proof of Proposition 1, $\mathrm{CVaR}_\alpha = 1 + 2c\varphi(c)/\alpha = 2[c\varphi(c) + \alpha/2]/\alpha$ with $c = z_\alpha$, which is 5.58 at $\alpha = 0.05$. The difference is $1 - 4.00/5.58 = 28\%$ of the one-axis value. Figure 1(c) shows the full curve for four values of α, computed by Monte Carlo with 6×10^5 draws.

Proposition 2(iii). Suppose $L_u \ge_{\mathrm{st}} L_v$ with $\mathbb{E}L_u = \mathbb{E}L_v$. Realize them with $L_u \ge L_v$ pointwise; then $L_u - L_v \ge 0$ has zero mean, so $L_u = L_v$ almost surely, and the two laws coincide. For two weight pairs with equal sum and different values, $\mathrm{Var}\, L = 2(\lambda_1^2 + \lambda_2^2)$ differs (since $\mathrm{Var}\, Z^2 = 2$), so the laws differ and neither dominates. *Sufficient condition.* If the sorted weights satisfy $\lambda_{(1)} \le \mu_{(1)}$ and $\lambda_{(2)} \le \mu_{(2)}$, then (relabelling the independent Z_i) $\lambda_{(1)} Z_1^2 + \lambda_{(2)} Z_2^2 \le \mu_{(1)} Z_1^2 + \mu_{(2)} Z_2^2$ pointwise, hence dominance. A family with a common ratio $\lambda_2/\lambda_1 = r$ is $\lambda = s(1, r)$, totally ordered by s, so it satisfies the condition. *Not necessary.* The pairs $(2, 1)$ and $(1, 0.9)$ have different ratios (0.5 and 0.9) but are componentwise ordered, so $(1, 0.9)$ is dominated by $(2, 1)$. An earlier draft of this paper stated that a common ratio was also necessary; this example shows it is not. $\square$

C. The effective dimension

Parts (i) and (ii). Let $X \sim \chi^2_m$ and $q = \chi^2_{m,1-\alpha}$. The size-biasing identity for the chi-square distribution,

$$\mathbb{E}\big[X\,\mathbf{1}\{X > q\}\big] = m\,P\big(\chi^2_{m+2} > q\big), \tag{S1}$$

follows from $x\, f_m(x) = m\, f_{m+2}(x)$ for the densities f_m, which is immediate from $f_m(x) = x^{m/2-1}e^{-x/2}/(2^{m/2}\Gamma(m/2))$ and $\Gamma(m/2+1) = (m/2)\Gamma(m/2)$. Since $\mathbb{E}X = m$ and X is continuous,

$$\frac{\mathrm{CVaR}_\alpha(X)}{\mathbb{E}X} = \frac{\mathbb{E}[X\mathbf{1}\{X > q\}]}{\alpha\,\mathbb{E}X} = \frac{1}{\alpha} P\big(\chi^2_{m+2} > q\big) = \rho_\alpha(m), \tag{S2}$$

and the ratio is invariant under the rescaling $L = (\sum_i w_i/m)X$ that turns equal weights into χ^2_m, giving (i).

For $m = 1$, $q = z_\alpha^2$ with $z_\alpha = \Phi^{-1}(1-\alpha/2)$, and $P(\chi^2_3 > z_\alpha^2) = \alpha + 2z_\alpha\varphi(z_\alpha)$ by integrating the χ^2_3 density by parts; dividing by α gives $\rho_\alpha(1) = 1 + 2z_\alpha\varphi(z_\alpha)/\alpha = r_\alpha$, i.e. Proposition 1. For $m = 2$, $X/2$ is exponential with unit mean, $q/2 = \ln(1/\alpha)$, and the memorylessness of the exponential gives $\mathbb{E}[X/2 \mid X/2 > \ln(1/\alpha)] = 1 + \ln(1/\alpha)$, i.e. $\rho_\alpha(2)$ is the equal-split value of Proposition 2. The two propositions of the main text are thus the $m = 1$ and $m = 2$ points of one curve.

Part (iii). χ^2_m/m has mean 1 and variance $2/m$ and concentrates at 1, so its α-tail average tends to 1; strict monotonicity in m follows because χ^2_m/m for $m' > m$ is a strict mean-preserving contraction of $\chi^2_{m'}/m'$ in the convex order, and CVaR_α is strictly monotone under that order for a continuous distribution [4]. Equation (6) extends to non-integer m by the same formula, and is smooth and decreasing there.

Part (iv): the participation-ratio approximation. For unequal weights we replace the spectrum $\{w_i\}$ by the equal-weight spectrum with the same first and second moments, which has exactly m_{eff} elements by Eq. (5); L and its surrogate then share their mean and variance, and the approximation amounts to matching the first two cumulants of the loss. It is not exact, because the tail of a weighted chi-square sum is eventually governed by the largest weight alone, so the error must grow as $\alpha \to 0$. We quantify it on 400 random spectra of 2–10 directions drawn in four styles (uniform weights; uniform cubed; log-uniform over four decades; and one dominant weight with small companions), each evaluated with 4×10^5 draws.

TABLE S1. Relative error of $\rho_\alpha(m_{\mathrm{eff}})$ against the true ratio over 400 random spectra. Negative means the approximation underestimates the ratio, i.e. is conservative for a tail objective.

α	0.20	0.10	0.05	0.025	0.01
median error (%)	0.42	-0.80	-1.94	-3.01	-4.24
95th pct. |error| (%)	1.16	2.28	5.29	7.78	10.37
worst error (%)	1.30	-3.07	-6.51	-9.53	-12.80
fraction conservative	0.22	0.97	0.99	0.99	0.99

At $\alpha = 0.05$ the approximation is conservative at 99% of spectra and within 6.5% everywhere; at $\alpha = 0.01$ the worst case reaches 12.8%. For the controls of Sec. S10, where $m_{\mathrm{eff}} \le 3$ and the spectra are mild, the observed error is a median 2.6%. Part (iv) is therefore an empirical statement with a stated domain, not a theorem, and the main text marks it as such.

D. The rare-channel mixture

Proposition 4, with the error term. Let $L = L_c + \ell_j \mathbf{1}_J$ with J independent of L_c, $P(J) = p < \alpha$, and let v_γ denote the $(1-\gamma)$-quantile of L_c. Assume $\ell_j > v_{\alpha-p}$, so that every point of J lies above every point of J^c that enters the tail.

Write $\gamma = \alpha - p$. The worst α of the mass of L is then exactly J together with the worst γ of J^c, and since CVaR at level α is the average of the worst α fraction,

$$\alpha\,\mathrm{CVaR}_\alpha(L) = p\big(\ell_j + \mathbb{E}[L_c \mid J]\big) + \gamma\,\mathrm{CVaR}_\gamma(L_c \mid J^c). \tag{S3}$$

Independence gives $\mathbb{E}[L_c \mid J] = \mathbb{E}L_c$ and $\mathrm{CVaR}_\gamma(L_c \mid J^c) = \mathrm{CVaR}_\gamma(L_c)$, so with $\beta = p\ell_j/\mathbb{E}L_c$ and $\mathbb{E}L = (1+\beta)\mathbb{E}L_c$,

$$\frac{\mathrm{CVaR}_\alpha(L)}{\mathbb{E}L} = \frac{1}{1+\beta}\left[\frac{\beta}{\alpha} + \frac{p}{\alpha} + \frac{\alpha-p}{\alpha}\,\frac{\mathrm{CVaR}_{\alpha-p}(L_c)}{\mathbb{E}L_c}\right], \tag{S4}$$

which is exact. Equation (8) is Eq. (S4) with $p/\alpha \to 0$. Two corrections are dropped. The term p/α is the jump branch's share of the *continuous* loss, a relative error of at most $(p/\alpha)/\rho_\alpha$. The second is the difference between $\frac{\alpha-p}{\alpha}\,\mathrm{CVaR}_{\alpha-p}(L_c)/\mathbb{E}L_c$ and ρ_α; since $\gamma\,\mathrm{CVaR}_\gamma$ is the tail integral, which is Lipschitz in γ with constant v_γ, that difference is at most $p\,v_{\alpha-p}/(\alpha\,\mathbb{E}L_c)$. Both are $O(p/\alpha)$ and both are positive, so Eq. (8) *underestimates* the ratio. Throughout Sec. S12, $p \le 2.0 \times 10^{-3}$ at $\alpha = 0.05$, so $p/\alpha \le 0.04$ and the bound on the relative error stays below 1.5% at every point — consistent with the observed median of 1.2% and with its sign.

Two limits and one warning. As $\beta \to 0$ the ratio returns to $\rho_\alpha(m_{\mathrm{eff}})$. As $\beta \to \infty$ it approaches $1/\alpha$, the largest ratio any loss can have when its whole tail mass sits in a single atom, so the mixture law interpolates between the continuous-noise prediction and that ceiling. The condition $p < \alpha$ matters: for $p > \alpha$ the worst α of the mass lies entirely inside J, the continuous part drops out of the tail, and the ratio saturates at $\ell_j/\mathbb{E}L$ instead. The evaluation grid of Sec. S12 deliberately spans $\alpha \in \{0.5, 0.2, 0.05, 0.01, 2\times10^{-3}, 5\times10^{-4}, 10^{-4}\}$ so that both sides of $p \approx \alpha$ are covered.

E. The reduced second-order model

Proposition 5.

Proof. (i) $\mathbb{E}L$ is quadratic in $\mathbf{e}_0$ with Hessian $2W$ and stationary at $\mathbf{e}_0 = -\mathbb{E}[A\mathbf{z} + Q(\mathbf{z})] = -\mathbb{E}\,Q(\mathbf{z})$. (ii) $\boldsymbol{\epsilon} = \|A\|\,(\hat{A}\mathbf{z} + \kappa\hat{Q}(\mathbf{z}) + \mathbf{e}_0/\|A\|)$; the factor $\|A\|^2$ cancels in the ratio defining R_α^{model}, and $\mathbf{e}_0/\|A\|$ ranges over $\mathbb{R}^3$ as $\mathbf{e}_0$ does. Scaling the noise by s sends $A \to sA$ and $Q \to s^2Q$. (iii) At $\kappa = 0$ the noise part $\hat{A}\mathbf{z}$ is symmetric under $\mathbf{z} \to -\mathbf{z}$, so $\text{CVaR}_\alpha(L(\mathbf{e}_0))$ is even in $\mathbf{e}_0$. It is also convex, since L is convex in $\mathbf{e}_0$ for every draw and CVaR is monotone and convex. Its minimum is therefore at $\mathbf{e}_0 = 0$, which is also the mean optimum. □

Two additional remarks:

(a) Why $\mathbf{e}_0/\|A\|$ is free. In (ii), the minimization of each objective runs over all $\mathbf{e}_0 \in \mathbb{R}^3$, and $\mathbf{e}_0 \mapsto \mathbf{e}_0/\|A\|$ is a bijection of $\mathbb{R}^3$, so the optimal *normalized* offset, and therefore both optimal values up to the common factor $\|A\|^2$, depend only on $(\kappa, \hat{A}, \hat{Q})$.

(b) What (ii) does not say. The scaling $\kappa \mapsto s\kappa$ holds for a *fixed* A and Q, i.e. a fixed control. If the control is re-optimized at each noise level, A and Q change as well. The prediction (c) of the prospective test (Sec. IV E) overlooked this, and failed for that reason.

F. Remark 1: the one-dimensional pre-compensation model

Let $\boldsymbol{\epsilon} = \mathbf{e}_0 + \mathbf{q}z^2$ and restrict the offset to the line $\mathbf{e}_0 = -c\,\mathbf{q}$ (components of $\mathbf{e}_0$ perpendicular to $\mathbf{q}$ only add a constant $|\mathbf{e}_\perp|_W^2 \geq 0$ to every draw and are optimally zero for both objectives, by translation invariance). With the loss metric W, $L = w\,(z^2 - c)^2$ with $w = \mathbf{q}^\mathsf{T}W\mathbf{q} > 0$, and w drops out of both argmins.

Mean. $\mathbb{E}(z^2 - c)^2 = \text{Var}(z^2) + (\mathbb{E}z^2 - c)^2 = 2 + (1 - c)^2$, minimized at $c = 1$ with value 2.

CVaR. By Eq. (1), $\text{CVaR}_\alpha L(c) = \min_\tau g(c, \tau)$, and g is jointly convex in (c, τ), so the optimum satisfies the first-order conditions. Differentiating under the expectation, $\partial_c g = \alpha^{-1}\mathbb{E}[\partial_c L\ \mathbf{1}\{L > \tau\}] = -2\alpha^{-1}\mathbb{E}[(z^2 - c)\mathbf{1}\{L > \tau\}]$, and at the optimal τ (the VaR) the condition $\partial_c g = 0$ reads

$$\mathbb{E}\big[z^2 - c \;\big|\; \text{tail}\big] = 0\,: \tag{S5}$$

the *average signed error over the tail draws vanishes.* The mean condition is the same with the whole distribution in place of the tail. Solving numerically (2×10^6 draws) gives $c = 2.76$ at $\alpha = 0.05$ (CVaR = 13.8, against 24.2 at $c = 1$) and $c = 1.70$ at $\alpha = 0.2$. The structure of the solution is instructive. At the optimum the VaR equals almost exactly c^2, the loss of a draw with $z = 0$, so the tail contains two kinds of draws: the *largest* draws, with $z^2 > 2c$, whose error is under-compensated (1.9% of all draws at $\alpha = 0.05$), and the *smallest* draws, with $z \approx 0$, whose error is over-compensated (the remaining 3.1%; 62% of the tail). Condition (S5) balances the two. The CVaR optimum is therefore the offset at which the worst over-compensated and the worst under-compensated draws are equally bad, which is why it lies much further out than the typical draw.

G. The κ^2 law

Throughout this section the reduced model of Proposition 5 is used in its normalized form: with $a = \|A\|_F$, $\hat{A} = A/a$ and $\hat{Q} = Q/(a\kappa)$,

$$\boldsymbol{\epsilon} = a\big(\hat{A}\mathbf{z} + \kappa\hat{Q}(\mathbf{z}) + \hat{\mathbf{e}}_0\big), \qquad L = \boldsymbol{\epsilon}^\mathsf{T}W\boldsymbol{\epsilon},$$

and the factor a^2 cancels in R_α. We drop the hats and write the offset as $\mathbf{e}_0 = \kappa\mathbf{f}$, which is the scale at which both optima live: the mean optimum is exactly $\mathbf{e}_0 = -\kappa\,\mathbb{E}\,Q(\mathbf{z})$.

Expansion of the loss. With $L_0 = \mathbf{z}^\mathsf{T}M\mathbf{z}$, $M = A^\mathsf{T}WA$,

$$L = L_0 + \kappa\,h_\mathbf{f}(\mathbf{z}) + \kappa^2 g_\mathbf{f}(\mathbf{z}), \qquad \begin{aligned} h_\mathbf{f} &= 2(A\mathbf{z})^\mathsf{T}W\big(Q(\mathbf{z}) + \mathbf{f}\big), \\ g_\mathbf{f} &= \big(Q + \mathbf{f}\big)^\mathsf{T}W\big(Q + \mathbf{f}\big). \end{aligned} \tag{S6}$$

L_0 and $g_\mathbf{f}$ are even in $\mathbf{z}$; $h_\mathbf{f}$ is odd, being a product of a linear and a quadratic form.

First order vanishes. Let $\tau^*(\kappa)$ attain the minimum in Eq. (1). Since the objective $\tau + \alpha^{-1}\mathbb{E}(L - \tau)_+$ is jointly convex and differentiable in κ for fixed τ, the envelope theorem gives

$$\frac{d}{d\kappa}\text{CVaR}_\alpha(L) = \frac{1}{\alpha}\,\mathbb{E}\big[(h_\mathbf{f} + 2\kappa g_\mathbf{f})\,\mathbf{1}\{L > \tau^*\}\big]. \tag{S7}$$

At $\kappa = 0$ the tail event is $\{L_0 > \tau_0\}$, which is invariant under $\mathbf{z} \to -\mathbf{z}$, while $h_\mathbf{f}$ is odd; the expectation vanishes. *The α-tail of a symmetric loss has two sides, and to first order a coherent offset gains on one exactly what it loses on the other.* This cancellation, and not any smallness of the offset itself, is why a threshold in κ exists.

Second order. Differentiating Eq. (S7) once more at $\kappa = 0$, using the value-at-risk sensitivity $d\tau^*/d\kappa = \mathbb{E}[h_{\mathbf{f}} \mid L_0 = \tau_0]$ (valid when L_0 has a continuous positive density ϱ at τ_0, which holds for a nondegenerate Gaussian quadratic form), the level-set term contributes the conditional variance of $h_{\mathbf{f}}$ and

$$\frac{d^2}{d\kappa^2}\mathrm{CVaR}_\alpha(L)\Big|_0 = \frac{2}{\alpha}\mathbb{E}\big[g_{\mathbf{f}}\mathbf{1}\{L_0 > \tau_0\}\big] + \frac{\varrho(\tau_0)}{\alpha}\mathrm{Var}\big[h_{\mathbf{f}} \mid L_0 = \tau_0\big], \tag{S8}$$

which is Eq. (12) after division by two. Both terms are quadratic in $\mathbf{f}$, with Hessian

$$H = 2W + \frac{4\varrho(\tau_0)}{\alpha}\,\Sigma_u, \qquad \Sigma_u = \mathrm{Cov}\big[WA\mathbf{z} \mid L_0 = \tau_0\big] \succeq 0, \tag{S9}$$

so J is strictly convex ($W \succ 0$) and its minimizer is explicit, $\mathbf{f}^\star = -H^{-1}\mathbf{b}$ with $\mathbf{b} = 2W\,\mathbb{E}[Q \mid L_0 > \tau_0]\,\alpha/\alpha + (2\varrho/\alpha)\mathrm{Cov}[h_0, WA\mathbf{z} \mid L_0 = \tau_0]$ and $h_0 = h_{\mathbf{0}}$. Since J is quadratic,

$$J(\mathbf{f}_{\mathrm{mean}}) - \min_{\mathbf{f}} J = \tfrac{1}{2}(\mathbf{f}_{\mathrm{mean}} - \mathbf{f}^\star)^{\mathsf{T}} H (\mathbf{f}_{\mathrm{mean}} - \mathbf{f}^\star), \tag{S10}$$

and dividing by $\mathrm{CVaR}_\alpha(L_0)$ gives Eq. (13). Both optima are $O(\kappa)$, so the remainder is $O(\kappa^3)$.

One dominant direction. Let $n = 1$ and let the second-order error be parallel to the first-order one, $\boldsymbol{\epsilon} = (e_0 + z + \kappa z^2)\mathbf{d}$. Then W enters only through $\mathbf{d}^{\mathsf{T}}W\mathbf{d} > 0$, which cancels in R_α, and with $e_0 = -\kappa c$ the loss is $L = F(z)^2$, $F(z) = z + \kappa(z^2 - c)$. Write $s = z_\alpha$, so that $2\big(1 - \Phi(s)\big) = \alpha$. The tail of L is $\{z > z_+\} \cup \{z < z_-\}$ with $F(z_\pm) = \pm V$, and expanding $z_+ = s + \kappa a_1 + \kappa^2 a_2$, $z_- = -s + \kappa b_1 + \kappa^2 b_2$, $V = s + \kappa v_1 + \kappa^2 v_2$ in Eq. (S6) gives

$$a_1 + s^2 - c = v_1, \qquad b_1 + s^2 - c = -v_1,$$

while the tail probability constraint $\big[1 - \Phi(z_+)\big] + \Phi(z_-) = \alpha$ reads, to first order, $-a_1 + b_1 = 0$. Hence $v_1 = 0$ and

$$a_1 = b_1 = c - s^2 : \tag{S11}$$

the two ends of the tail move by the same amount and in the same direction. At second order the same constraint gives $b_2 - a_2 = -s\,a_1^2$. Collecting the three contributions to $\alpha\mathrm{CVaR}_\alpha(L)$ — the integral of $F^2\varphi$ over the symmetric region $|z| > s$, the boundary shifts, and the curvature of $F^2\varphi$ at $\pm s$ — the terms in s^2 cancel and

$$\alpha\,\mathrm{CVaR}_\alpha(L) = \underbrace{\mathbb{E}\big[z^2\mathbf{1}_{|z|>s}\big]}_{= 2s\varphi(s) + \alpha\, =\, \alpha r_\alpha} + \kappa^2\Big[m_2(c) + 2s\varphi(s)\,(c - s^2)^2\Big] + O(\kappa^3), \tag{S12}$$

with $m_2(c) = \mathbb{E}[(z^2 - c)^2\mathbf{1}_{|z|>s}]$. The bracket is a quadratic in c with leading coefficient $\alpha + 2s\varphi(s) = \alpha r_\alpha$ and stationary point

$$c^\star = \frac{\mathbb{E}[z^2\mathbf{1}_{|z|>s}] + 2s^3\varphi(s)}{\alpha + 2s\varphi(s)} = 1 + s^2\Big(1 - \frac{1}{r_\alpha}\Big),$$

using $2s\varphi(s) = \alpha(r_\alpha - 1)$. Therefore $\mathrm{CVaR}_\alpha(L) = r_\alpha\big[1 + \kappa^2(c - c^\star)^2\big] + \mathrm{const} + O(\kappa^3)$, and since the mean-optimal offset is $c = 1$,

$$R_\alpha(\kappa) = \frac{\mathrm{CVaR}_\alpha(L)|_{c=1} - \mathrm{CVaR}_\alpha(L)|_{c=c^\star}}{\mathrm{CVaR}_\alpha(L)|_{c=1}} = (c^\star - 1)^2\kappa^2 + O(\kappa^3), \tag{S13}$$

the factor r_α cancelling. This is Eq. (14).

The two limits of the offset. $c^\star_\alpha \to 1 + z_\alpha^2(1 - 1/r_\alpha)$ as $\kappa \to 0$ and $c \to 2.76$ ($\alpha = 0.05$) or 1.70 ($\alpha = 0.2$) as $\kappa \to \infty$ (Remark 1), so the optimal pre-compensation *decreases* with κ: $4.15 \to 2.76$ at $\alpha = 0.05$. Direct minimization of the scalar model reproduces this, $c^\star = 4.20, 4.17, 3.82, 3.16, 2.86$ at $\kappa = 0.02, 0.1, 0.5, 1.2, 2.2$.

Numerical verification. `make_kappa_bound.py` (Sec. S15) checks every step without the simulator: (i) in the scalar model the measured R_α/κ^2 is 9.69 and 1.29 at $\kappa = 0.05$ against the predicted 9.94 and 1.29 for $\alpha = 0.05$ and 0.2, and the measured $c^\star$ is 4.14 and 2.14 against 4.15 and 2.14; (ii) for random three-dimensional geometries the quadratic form $J(\mathbf{f})$ of Eq. (12) agrees with direct Monte Carlo evaluation of $[\mathrm{CVaR}_\alpha(\kappa, \mathbf{f}) - \mathrm{CVaR}_\alpha(0)]/\kappa^2$ to about 1% at $\kappa = 0.02$–0.04; (iii) applied to the fitted A and Q of every diagnosed mean control, the closed form reproduces the numerically optimized reduced model with median ratio 1.07 (interquartile range 0.94–1.12) over the 30 controls with $\kappa < 0.25$ [Fig. 8(b)].

Quantity	Value	Note
transmon levels N	4 (or 2)	2 levels: leakage ablation
anharmonicity $\alpha_{\rm anh}/2\pi$	-200 MHz	typical fixed-frequency transmon [6, 7]
qubit frequency	5 GHz	enters only the thermal population
temperature	50 mK	thermal occupation $n_{\rm th} \approx 0.0083$
T_1, T_2^*	45.5 μs, 16.8 μs	$\gamma_\varphi = 1/T_2^* - 1/(2T_1)$
target gate	$X_{\pi/2} = (I - iX)/\sqrt{2}$	on the computational subspace
gate time T_g	6.3–32.5 ns	baseline 12.7 ns (leakage onset)
time step	0.25 ns, $N_t \geq 40$	piecewise-constant propagation
basis modes	$\min(40, \lfloor N_t/3 \rfloor)$ per quadrature	17 modes per quadrature at 12.7 ns ($N_t = 51$)
amplitude bound A	$2.5\,\pi/T_g$	2.5× peak of a Hann pulse of equal area
filter	Butterworth, order 4, 300 MHz	zero-phase (forward–backward)
leakage weight λ	7	in $L = 1 - \bar{F} + \lambda P_{\rm leak}$
detuning noise $\sigma_\delta/2\pi$	2.5 MHz (0.5–25 MHz in sweeps)	quasi-static
amplitude noise σ_γ	2% (1–12.7% in Stage 1)	quasi-static
correlation ρ	0 (-0.99 to 0.99 in Stage 1)	between δ and γ

TABLE S2. Model and simulation parameters.

S2. MODEL, SIMULATORS AND OPTIMIZATION

Hamiltonian and dissipation. The Hamiltonian is Eq. (15), with parameters in Table S2. With $N = 4$ the model contains the two leakage levels a fast pulse can reach; the two-level model is the same code with the matrices truncated, so that leakage is impossible by construction. Dissipation enters through

$$\dot{\varrho} = -i[H, \varrho] + \sum_k \Big(C_k \varrho C_k^\dagger - \tfrac{1}{2}\{C_k^\dagger C_k, \varrho\} \Big), \qquad C_1 = \sqrt{\gamma_1}\, a, \;\; C_2 = \sqrt{2\gamma_\varphi}\, a^\dagger a, \tag{S14}$$

with $\gamma_1 = 1/T_1$ and $\gamma_\varphi = 1/T_2^* - 1/(2T_1)$. A 12.7 ns gate loses 10^{-4}–10^{-3} of fidelity to these processes, almost independently of the control and identically for every quasi-static draw. On the qubit, δ adds $\frac{\delta}{2}Z$ (a rotation about z) and γ scales the drive (a rotation about the drive axis), so the two noise sources rotate the qubit about different axes.

Pulse parameterization. Each quadrature is a sum of Hann-windowed sine modes chosen so that the pulse starts and ends at zero:

$$\tilde{u}_I(t) = \sum_{n=1}^{M} c_n^I \, \sin\frac{(2n-1)\pi t}{T_g}\, h(t), \qquad \tilde{u}_Q(t) = \sum_{n=1}^{M} c_n^Q \, \sin\frac{2n\pi t}{T_g}\, h(t), \tag{S15}$$

with the Hann window $h(t) = \frac{1}{2}(1 - \cos 2\pi t/T_g)$. The in-phase modes are symmetric about $T_g/2$ and the quadrature modes antisymmetric, the natural symmetry of a DRAG-type correction [8], although the optimizer is free to combine them arbitrarily. The raw envelope is bounded smoothly,

$$u = A \tanh(\tilde{u}/A), \tag{S16}$$

which keeps the pulse inside the amplitude budget $\pm A$ with an exact chain rule. An earlier version clipped the pulse and back-propagated as if it had not (pitfall 4 in Sec. S3.D). Because $A = 2.5\,\pi/T_g$ scales with the gate time, the budget is equally tight at every T_g and the only physical change along a gate-time sweep is the gate time itself. The optimized controls use the whole budget: at 12.7 ns the raw envelopes of both quadratures saturate at $\pm A/2\pi = \pm 98$ MHz, and the filtered pulses that reach the qubit peak at about 115 MHz because of the filter's overshoot (Fig. S5).

Bandwidth filter. Real arbitrary-waveform generators and cabling limit the bandwidth. The bounded envelope is passed through a 4th-order Butterworth low-pass filter with a 300 MHz cutoff, applied forward and backward (zero phase, Gustafsson initial conditions). Because the filter is linear in the samples, we build it once as an exact $N_t \times N_t$ matrix F; the simulated pulse is $u_f = Fu$ and the gradient with respect to u is $F^\top \partial L/\partial u_f$.

Loss. For each noise draw the loss is Eq. the loss of Sec. III, evaluated on four input states. Each input is a mixture of the ideal pure state and the thermal population of the leakage levels, $\varrho_{\text{in}} = (1 - p_{\text{bg}})|\psi\rangle\langle\psi| + P\varrho_{\text{th}}P$, with P the projector onto levels ≥ 2 and $p_{\text{bg}} = 6.8 \times 10^{-5}$, so that a perfect gate still shows the thermal leakage background. The per-state figure of merit is $J_\psi = \text{Re}\,\text{Tr}\big[(\varrho_{\text{tgt}} - \lambda P)\,\varrho_{\text{out}}\big]$ with $\varrho_{\text{tgt}} = U_{\text{target}}\varrho_{\text{in}}U_{\text{target}}^\dagger$, and $L = 1 - \frac{1}{4}\sum_\psi J_\psi$. This is linear in the output state, which makes the gradient an adjoint (backward) propagation.

Noise ensembles. Each draw is a pair (δ, γ). Three disjoint ensembles share one distribution: training ($K = 128$, seed 11), validation (300, seed 22) and test (1000, seed 33). A replication seed s shifts all three seeds by $1000s$. Correlated channels are generated from independent normals δ and g as

$$\gamma = \rho\,\sigma_\gamma\,\frac{\delta}{\sigma_\delta} + \sqrt{1-\rho^2}\,g, \qquad g \sim \mathcal{N}(0, \sigma_\gamma^2), \tag{S17}$$

which gives $\text{corr}(\delta, \gamma) = \rho$ with unchanged marginals and reproduces the uncorrelated draws exactly at $\rho = 0$, so every Stage-0 point remains valid. Using one distribution for all three ensembles matters: an earlier version trained on wider noise than it tested on, which by itself made CVaR controls look worse (pitfall 2).

A. Two simulators

Full Lindblad model. The reference simulator propagates the density matrix under Eq. (S14) with piecewise-constant controls. On each time step the Liouvillian $\mathcal{L}_k$ (a $N^2 \times N^2$ matrix, 16×16 for 4 levels) is exponentiated exactly. The gradient with respect to the control on step k needs the derivative of $e^{\mathcal{L}_k\Delta t}$ in the direction $\partial\mathcal{L}_k/\partial u$, which we obtain exactly from the block-matrix identity [9]

$$\exp\begin{pmatrix}\mathcal{L}_k & \partial\mathcal{L}_k \\ 0 & \mathcal{L}_k\end{pmatrix}\Delta t = \begin{pmatrix} e^{\mathcal{L}_k\Delta t} & \int_0^{\Delta t} e^{\mathcal{L}_k(\Delta t - s)}\,\partial\mathcal{L}_k\,e^{\mathcal{L}_k s}\,ds \\ 0 & e^{\mathcal{L}_k\Delta t}\end{pmatrix}. \tag{S18}$$

Combined with a forward pass for the state and a backward pass for the adjoint of J, this gives the exact gradient at the cost of about three simulations. Per-draw evaluations are distributed over worker processes, and draw i is always returned at row i, which keeps the bootstrap paired (pitfall 3).

Fast closed-system model. Optimizing with the full model costs minutes per run. T_1 and T_2^* act nearly identically on every quasi-static draw, so to a good approximation they add a constant to each per-draw loss, and $\text{CVaR}_\alpha(L+c) = \text{CVaR}_\alpha(L)+c$ does not move the optimum. We therefore *optimize* on a closed-system model in which the Hamiltonians of all K draws and all N_t time steps are diagonalized in one vectorized call, $H_{k,j} = V\,\text{diag}(e)\,V^\dagger$, so each step propagator is $V\,\text{diag}(e^{-ie\Delta t})V^\dagger$. The derivative of the propagator in direction X follows from the Daleckii–Krein formula [10]: in the eigenbasis,

$$\big[D\,e^{-iH\Delta t}[X]\big]_{mn} = G_{mn}\,(V^\dagger X V)_{mn}, \qquad G_{mn} = \begin{cases} \dfrac{e^{-ie_m\Delta t} - e^{-ie_n\Delta t}}{e_m - e_n}, & e_m \neq e_n, \\ -i\Delta t\,e^{-ie_m\Delta t}, & e_m = e_n. \end{cases} \tag{S19}$$

With dissipation switched off, the fast model reproduces the Lindblad code to 10^{-7} in both loss and gradient (self-test). It is fast enough to run a full multistart point in about a minute.

Every reported number is a full-Lindblad re-evaluation. After optimization, all saved controls (mean, CVaR at each α, reverse starts) are re-simulated in the full Lindblad model on the test ensemble, and all statistics are recomputed there. How well the two simulators agree is quantified in Sec. S5.

B. Objectives and optimizer

All objectives share one code path. For a control with parameters $\mathbf{c}$, the training losses $L_k(\mathbf{c})$ and their gradients are computed for all K draws and combined with weights w_k:

$$\mathcal{J}(\mathbf{c}) = \sum_{k=1}^{K} w_k\,L_k(\mathbf{c}), \qquad \nabla\mathcal{J} = \sum_{k=1}^{K} w_k\,\nabla L_k(\mathbf{c}). \tag{S20}$$

The mean uses $w_k = 1/K$. CVaR uses the exact Rockafellar–Uryasev weights of Eq. (1): sort the draws by loss, give the worst $\lfloor\alpha K\rfloor$ weight $1/(\alpha K)$ and the next one the fractional remainder. At $\alpha = 0.05$ and $K = 128$ that is six draws at weight 0.156 and one at 0.0625. Where two draws tie at the boundary the weights are a valid subgradient, and ties do not occur in practice with continuous noise.

Optimizer. L-BFGS-B [11] with at most 500 iterations (1000 in the strong search), ftol $= 10^{-12}$ and gtol $= 10^{-8}$, identical for every objective. The CVaR objective is only piecewise smooth, because the set of tail draws changes as the control moves; in practice L-BFGS-B handles this well, and the in-sample audit (below) checks that it actually decreases the training CVaR.

Restarts. Each objective gets the same number of restarts, 4 by default (20 in the strong search). The list is: the Hann pulse of the correct area; random perturbations of it (independent Gaussian noise of scale $0.3A$ on every sample, projected back onto the basis); and, along a sweep, a continuation from the previous point's mean control, rescaled to keep the rotation area. For CVaR, one restart is replaced by the warm start from u_{mean}.

C. Extracting the error rotation vector

Given the 2×2 qubit block U_q of a propagator, form $E = U_{\mathrm{target}}^{\dagger} U_q$. If E is not unitary (leakage), replace it by its polar factor $E \to UV^{\dagger}$ from the singular value decomposition $E = U\Sigma V^{\dagger}$, the closest unitary in Frobenius norm. Divide by $\sqrt{\det E}$ to land in $SU(2)$, where $E = c\, I - i\, \mathbf{v} \cdot \boldsymbol{\sigma}$ with $c = \frac{1}{2}\mathrm{Re}\,\mathrm{Tr}\, E$ and $v_j = -\frac{1}{2}\mathrm{Im}\,\mathrm{Tr}(E\sigma_j)$. The square root leaves a sign ambiguity $E \leftrightarrow -E$, resolved by choosing $c \geq 0$, i.e. a rotation angle $\epsilon \leq \pi$. Then $\epsilon = 2\,\mathrm{atan2}(|\mathbf{v}|, c)$ and $\boldsymbol{\epsilon} = \epsilon\, \mathbf{v}/|\mathbf{v}|$.

D. Exact gradient of the closed-system loss

For one draw and one input, $J = \mathrm{Re}\,\mathrm{Tr}[O\, \varrho_N]$ with $\varrho_{k+1} = U_k \varrho_k U_k^{\dagger}$. Define the backward-propagated observable $\Lambda_N = O$, $\Lambda_k = U_k^{\dagger} \Lambda_{k+1} U_k$. A change dU_k of one step propagator changes J by $dJ = 2\,\mathrm{Re}\,\mathrm{Tr}\big[\Lambda_{k+1}\, dU_k\, \varrho_k U_k^{\dagger}\big]$ (for Hermitian O). With $M_k = \varrho_k U_k^{\dagger} \Lambda_{k+1}$ and $dU_k = V\big(G \circ (V^{\dagger} X V)\big) V^{\dagger}\, du$ from Eq. (S19),

$$\frac{\partial J}{\partial u_k} = 2\,\mathrm{Re} \sum_{m,n} G_{mn}\, (V^{\dagger} X V)_{mn}\, (V^{\dagger} M_k V)_{nm},$$

with $X = (1+\gamma)(a + a^{\dagger})/2$ for u_I and $(1+\gamma)\, i(a^{\dagger} - a)/2$ for u_Q. One forward pass stores the ϱ_k, one backward pass builds the Λ_k, and the cost is independent of the number of control parameters. The gradient with respect to the basis coefficients follows by the chain rule through the filter F and the bound (S16), $\partial u / \partial \tilde{u} = 1 - (u/A)^2$.

E. The four-state loss metric

Let the error $E = U_{\mathrm{target}}^{\dagger} U = \cos(\epsilon/2) I - i \sin(\epsilon/2)\, \mathbf{n} \cdot \boldsymbol{\sigma}$ act on an input with Bloch vector $\mathbf{r}$. The state fidelity is $F = |\langle\psi|E|\psi\rangle|^2 = |\cos(\epsilon/2) - i \sin(\epsilon/2)\, \mathbf{n} \cdot \mathbf{r}|^2 = \cos^2(\epsilon/2) + \sin^2(\epsilon/2)(\mathbf{n} \cdot \mathbf{r})^2$, so

$$1 - F_{\mathbf{r}} = \sin^2(\epsilon/2)\big(1 - (\mathbf{n} \cdot \mathbf{r})^2\big).$$

The four inputs $|0\rangle, |1\rangle, |+\rangle, |+i\rangle$ have $\mathbf{r} = \hat{z}, -\hat{z}, \hat{x}, \hat{y}$, and $\sum_{\mathbf{r}} (\mathbf{n} \cdot \mathbf{r})^2 = 2n_z^2 + n_x^2 + n_y^2 = 1 + n_z^2$. Hence

$$1 - \bar{F} = \tfrac{1}{4} \sin^2(\epsilon/2)\, (3 - n_z^2) \approx \tfrac{1}{16}\big(3\epsilon_x^2 + 3\epsilon_y^2 + 2\epsilon_z^2\big) = \boldsymbol{\epsilon}^{\mathsf{T}} W \boldsymbol{\epsilon}, \qquad W = \tfrac{1}{16}\mathrm{diag}(3, 3, 2).$$

We checked this expression against the four-state loss computed by the closed-system simulator for the 2-level model. On the transmon the thermal admixture of the inputs and the leakage term add small corrections that W does not include.

S3. FAIRNESS PROTOCOL, STATISTICS AND PIPELINE ERRORS

A. The fairness protocol

A small difference between two optimized controls is easily created by the optimization procedure rather than by the objectives. The protocol below was built up in response to specific failures (Sec. S3.D).

1. *Equal restarts and tolerances* for every objective.

2. *Warm start.* One CVaR restart begins at the selected mean control. L-BFGS-B only accepts steps that decrease the objective, so this run *cannot* end with a higher training CVaR than u_{mean}. If it does, the code is wrong. This *in-sample audit* is checked at every point.

3. *Budget parity.* The warm-started CVaR run has received the mean optimization plus one more run. The mean objective therefore gets one extra continuation run from its own optimum, kept if it improves the validation mean. This removes the possibility that CVaR "wins" only because it was optimized for longer.

4. *Selection on validation data only.* Each objective picks its winning restart by its own risk measure on the validation ensemble. The test ensemble is used only for the reported numbers.

5. *Reverse warm start.* For each CVaR control, the mean optimizer is restarted from u_{CVaR}. The result is classified, at a relative tolerance of 10^{-3} in validation mean loss, as: *same basin* (it walks back to the mean loss of u_{mean}, so CVaR shifted the solution within one basin); *different basins* (it stays at a worse mean loss); or *artifact* (it finds a better mean control than the multistart did, so the multistart had missed a basin).

6. *Conservative comparator.* The reported R_{tail} compares u_{CVaR} with the best mean-objective control found at that point by any route (multistart, continuation or reverse start), chosen on validation data. The mean side thus receives more search than the CVaR side. A positive R_{tail} is a *lower bound* on the CVaR advantage. The converse does not hold: if a reverse start upgraded the comparator by more than 0.1%, a negative R_{tail} may just mean that the CVaR search did not get the same upgrade. Such cells are drawn with open markers, their same-budget value is listed in Sec. S14, and no cell is dropped.

B. Evaluation and statistics

The outcome metrics, on the test ensemble in the full Lindblad model, are the tail gap and the relative tail gain,

$$D(\alpha) = \mathrm{CVaR}_\alpha(L[u_{\mathrm{mean}}]) - \mathrm{CVaR}_\alpha(L[u_{\mathrm{CVaR}}]), \qquad R_{\mathrm{tail}}(\alpha) = \frac{D(\alpha)}{\mathrm{CVaR}_\alpha(L[u_{\mathrm{mean}}])}. \tag{S21}$$

$R_{\mathrm{tail}} > 0$ means the CVaR control has the better tail; $R_{\mathrm{tail}} = 0.2$ means its worst-5% average error is 20% lower. Uncertainty comes from a *paired* bootstrap [12]: the 1000 test draws are resampled with replacement 2000 times, the *same* resampled indices are used for both controls, and D and R_{tail} are recomputed each time; the 95% interval is the 2.5–97.5 percentile range. Pairing matters because the two controls are evaluated on the same draws, and much of the draw-to-draw variation is common to both; an unpaired bootstrap would give much wider intervals. A point counts as a significant divergence when the interval excludes zero, and as within noise when it contains zero. The *out-of-sample audit* for the warm-started CVaR control requires $D \geq 0$ up to its interval.

Test-set resolution. With 1000 test draws, $\mathrm{CVaR}_{0.05}$ averages 50 draws, which resolves differences of a few percent. $\mathrm{CVaR}_{0.01}$ averages 10 test draws but was optimized on 1.28 training draws, which is why that level overfits (Sec. S3.C). Replication across independent noise seeds shows that out-of-sample R_{tail} scatters by about ± 0.04 at points where the objectives are equivalent, which sets the practical resolution of the study.

C. Audit and replication

Before any physics, the pipeline must pass its own test: a CVaR optimizer started from u_{mean} must not end with a worse training CVaR, and should not end with a significantly worse test CVaR either.

α	0.01	0.025	0.05	0.10	0.20
seed 0: R_{tail}	-0.038	-0.000	-0.010	$+0.006$	-0.004
95% CI	$[-0.147, 0.062]$	$[-0.012, 0.012]$	$[-0.031, 0.008]$	$[-0.005, 0.018]$	$[-0.016, 0.009]$
seed 1: R_{tail}	-0.137	-0.041	-0.034	-0.038	-0.010
seed 2: R_{tail}	$+0.016$	$+0.013$	$+0.018$	$+0.005$	-0.001

TABLE S3. E1 and E6, $T_g = 12.7$ ns, full Lindblad, test ensemble. The in-sample audit passes for every seed and α. As the audit requires, R_{tail} here compares the CVaR control with the mean control it was warm-started from (equal budget). Out of sample, R_{tail} scatters across seeds by about ± 0.04 around zero; with seed 1 the CVaR control is significantly *worse* at every α.

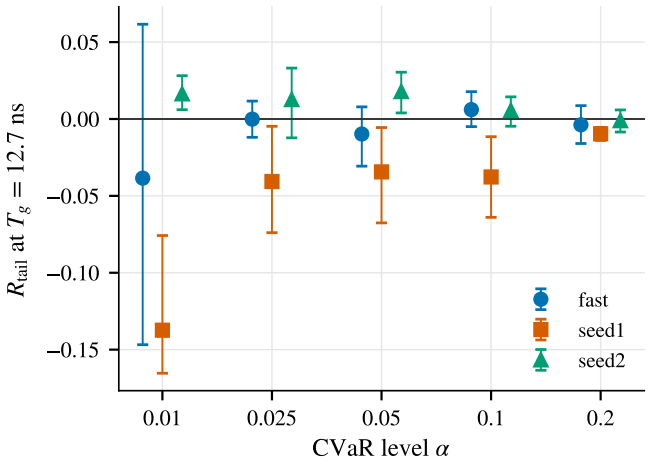


FIG. S1. Audit (E1, E6): R_{tail} at the leakage onset, $T_g = 12.7$ ns, for five CVaR levels and three independent noise seeds ("fast" is the first seed, from the main run), full Lindblad, CVaR control against the mean control it was warm-started from. The scatter between seeds ($\approx \pm 0.04$) sets the resolution of the study. At $\alpha = 0.01$ the CVaR rests on 1.3 training draws and overfits.

All five CVaR levels pass the in-sample audit, and none fails out of sample (Table S3, Fig. S1). At $\alpha = 0.01$ the CVaR averages just $0.01 \times 128 \approx 1.3$ training draws; its training CVaR improves while its test CVaR does not, which is finite-sample overfitting rather than a physical effect. We therefore treat $\alpha = 0.01$ as below the resolution of $K = 128$.

The replication (E6) sharpens this. With three independent training, validation and test ensembles, R_{tail} at the onset ranges from -0.04 to $+0.02$ for $\alpha \geq 0.025$. There is no systematic CVaR advantage, and one seed shows a significant disadvantage. This is expected: CVaR at $\alpha = 0.05$ is estimated from about 6 of the 128 training draws, so the CVaR control fits its training sample more closely than the mean control does. Where the theorem says the two objectives share a minimizer, CVaR can only add estimation noise.

D. Seven ways to manufacture a spurious result

Each of the following occurred in an earlier version of this pipeline and was caught by the audit. They are listed because each would have produced a publishable-looking but false conclusion, and because most of them are generic to comparisons between optimization objectives.

	Error	Spurious effect	Cure
1	loss defined as $-(F - \lambda P_{\mathrm{leak}}) < 0$	sign of R_{tail} flipped; a worse control read as better	loss ≥ 0, larger is worse
2	training noise wider than test noise	CVaR pays for robustness never tested	one distribution for all ensembles
3	parallel results collected in completion order	"paired" bootstrap silently unpaired	draw i always at row i
4	hard clipping with unclipped gradient	wrong gradients on up to 58% of samples	smooth bound $A \tanh(\tilde{u}/A)$
5	auxiliary regularizer 1–5× the loss	$R_{\mathrm{tail}} \approx 0.3$ at the onset, ≈ 0 without	regularizer removed
6	mean optimizer stuck in a poor basin	any other objective looks good	reverse warm start, best-mean comparator
7	mean given extra search, CVaR not	CVaR looks arbitrarily bad ($R_{\mathrm{tail}} = -3.6$)	conservative-comparator rule

TABLE S4. Pipeline errors found and removed.

1. Defining the loss as $-(F - \lambda P_{\text{leak}}) < 0$ flips the sign of R_{tail}, because its denominator becomes negative: a CVaR control that was *worse* on every test draw read as "better".
2. Training on wider noise (2.5 MHz, 2%) than the test set (1 MHz, 1%) makes a CVaR control pay, in nominal fidelity, for robustness against tails that never occur in the test set.
3. Collecting parallel results in completion order silently unpairs a "paired" bootstrap: the same index then refers to different draws for the two controls.
4. Hard-clipping the pulse while back-propagating as if unclipped gives wrong gradients on up to 58% of samples, since that fraction of samples sat on the rail.
5. An auxiliary filter-function regularizer, sized as 2% of the fidelity *reward*, was 1–5× the infidelity *loss*. The optimizer then mostly minimized a noise model that was neither in the ensemble nor in the reported CVaR. It manufactured $R_{\text{tail}} \approx 0.3$ at the leakage onset; without it, $R_{\text{tail}} \approx 0$.
6. A mean optimizer stuck in a poor basin makes *any* other objective look good. The reverse warm start exposed 2–9% better mean controls at long gate times.
7. The converse. A mean control found with *extra* search, if CVaR is not given the same, can make CVaR look arbitrarily bad: in one cell ($T_g = 20$ ns, $\alpha = 0.2$), $R_{\text{tail}} = -3.6$ against a mean control found by a reverse start, whereas the same cell reads $+0.11$ at equal budget. Hence the conservative-comparator rule of Sec. S3: positive values are lower bounds, and negative values in upgraded cells are not interpreted.

Each has the same cure: an audit that could have made the headline result fail. Items 6 and 7 are two sides of one problem, the asymmetry of search effort between objectives, and they show that neither the equal-budget nor the best-found comparison is safe alone. Reporting both, as Sec. S14 does, is the honest option.

S4. ERROR-GEOMETRY DIAGNOSTICS

To understand *why* two controls differ, we extract the full error of every test draw, not only its loss.

Error rotation vectors. For each draw, the qubit block of the closed-system propagator is compared with the target, $E = U_{\text{target}}^{\dagger} U_{2\times 2}$. If leakage makes the block slightly non-unitary, it is replaced by its nearest unitary (the polar factor from a singular value decomposition). After removing the global phase, $E = \cos(\epsilon/2) I - i \sin(\epsilon/2)\, \mathbf{n} \cdot \boldsymbol{\sigma}$, and the error rotation vector $\boldsymbol{\epsilon} = \epsilon \mathbf{n}$ follows from the trace and the Pauli components of E, with the sign fixed so that $\epsilon \le \pi$. For a 2-level model this is exact; its four-state loss is $\boldsymbol{\epsilon}^{\mathsf{T}} W \boldsymbol{\epsilon}$ to leading order, which we verified against the simulator (Sec. S2.E).

Whitened noise coordinates. The two noise variables are standardized, $x_1 = \delta/\sigma_\delta$, $x_2 = \gamma/\sigma_\gamma$, and rotated into independent standard normals:

$$z_{\pm} = \frac{x_1 \pm x_2}{\sqrt{2(1 \pm \rho)}}, \tag{S22}$$

ordered so that the direction with the larger variance $1 + |\rho|$ comes first (z_+ for $\rho \ge 0$, z_- for $\rho < 0$). With one active channel, $z = x_1$ or x_2. Whitening keeps the fit well conditioned at $|\rho| = 0.99$, where x_1 and x_2 are almost collinear.

Expansion and κ. Over the 1000 test draws we fit, by least squares, each component of $\boldsymbol{\epsilon}$ as a quadratic polynomial in $\mathbf{z}$:

$$\boldsymbol{\epsilon}(\mathbf{z}) \approx \mathbf{e}_0 + \sum_i A_{\cdot i} z_i + \sum_{i \le j} Q_{\cdot ij} z_i z_j. \tag{S23}$$

The coherent error is $\mathbf{e}_0$, the first-order response A (a $3 \times n$ matrix) and the second-order coefficients Q (a $3 \times n(n+1)/2$ array). The mechanism variable is $\kappa = \|Q\|_F / \|A\|_F$, computed for the mean control. For one channel it reduces to $|\mathbf{q}|/|\mathbf{a}|$, where $\boldsymbol{\epsilon} \approx \mathbf{e}_0 + \mathbf{a} z + \mathbf{q} z^2$.

Reduced-model prediction. From the fitted A and Q of the mean control, $R_{0.05}^{\text{model}}$ (Proposition 5) is computed with no free parameter: we draw 6×10^4 standard normal vectors $\mathbf{z}$, form the noise part $A\mathbf{z} + Q(\mathbf{z})$, and minimize $\text{CVaR}_{0.05}$ of $L = (\mathbf{e}_0 + A\mathbf{z} + Q(\mathbf{z}))^{\mathsf{T}} W (\cdots)$ over $\mathbf{e}_0 \in \mathbb{R}^3$ by Nelder–Mead, starting from the exact mean optimum $\mathbf{e}_0 = -\mathbb{E}\, Q(\mathbf{z})$. Then $R^{\text{model}} = 1 - \text{CVaR}^{\min} / \text{CVaR}(\mathbf{e}_0^{\text{mean}})$.

Offsets. For points where one noise direction dominates, the coherent offset of each control is summarized as $c = -\mathbf{e}_0 \cdot \mathbf{q} / |\mathbf{q}|^2$, the offset in units of the second-order error, to be compared with Remark 1.

Loss-quadratic geometry. For Proposition 2, the per-draw loss is also fitted directly as $L \approx L_0 + \mathbf{g}^\mathsf{T}\mathbf{x} + \mathbf{x}^\mathsf{T} M \mathbf{x}$; the weights $\lambda_1 \geq \lambda_2$ are the eigenvalues of $C^{1/2} M C^{1/2}$ with C the channel correlation matrix, and the loss anisotropy is $(\lambda_1 - \lambda_2)/(\lambda_1 + \lambda_2)$. This fit is coarser than Eq. (S23), because the loss is quartic in the noise once Q matters: its R^2 ranges from 0.55 to 0.998 over the diagnosed mean controls.

S5. SIMULATOR VALIDATION

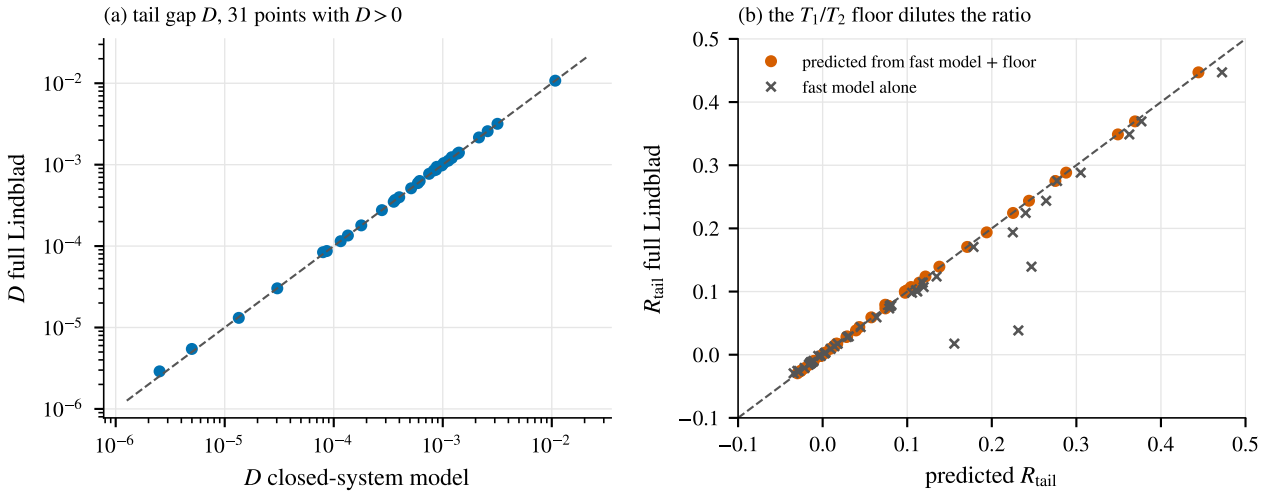


FIG. S2. How the fast optimizer model relates to the full Lindblad model, at all 44 points with saved error clouds. (a) The absolute tail gap D agrees across the two models over four decades (median ratio 1.000, interquartile range 0.998–1.022). (b) The ratio R_{tail} does not, at low losses: the Lindblad value (vertical) is predicted to within 0.005 by the fast D divided by the fast CVaR plus the T_1/T_2 floor (orange; median error 0.0003), while the fast R_{tail} alone (crosses) overstates it where the losses fall below the floor.

The fast model leaves out T_1 and T_2^*. To check that this does not bias the conclusions, we compared both models at every point for which closed-system error clouds were saved (44 points, Fig. S2). Two results emerge. First, the *absolute* gap D is the same in both models: the median ratio $D_{\mathrm{Lindblad}}/D_{\mathrm{fast}}$ is 1.000, with interquartile range 0.998–1.022. Dissipation adds almost the same constant to both controls' losses and cancels in the difference, as translation invariance of CVaR predicts. Second, the *relative* gain R_{tail} is diluted, because the constant enters its denominator:

$$R_{\mathrm{tail}}^{\mathrm{Lindblad}} \approx \frac{D_{\mathrm{fast}}}{\mathrm{CVaR}_\alpha^{\mathrm{fast}}(L[u_{\mathrm{mean}}]) + \ell_{\mathrm{floor}}}, \tag{S24}$$

where ℓ_{floor}, the mean-loss difference between the two models for u_{mean}, ranges from 1.4×10^{-4} to 1.06×10^{-3}. Equation (S24) predicts the Lindblad R_{tail} with median absolute error 0.0003 and maximum 0.005. The fast R_{tail} alone has median error 0.003 but reaches 0.19 at points whose losses lie well below the floor. All conclusions in this paper are drawn from the Lindblad values, which are the conservative ones; where the dilution matters (the prospective test at a quarter of the noise) both are reported.

S6. EXPERIMENTS

The experiments are listed in Table S5. The *leakage onset* used to place the E2 sweep is the gate time at which the nominal leakage of a Hann pulse exceeds 5× the thermal background, found by log–log interpolation on 30 gate times between 3 and 120 ns (Fig. S3). E4 matches $\sigma_\phi^2 = (\sigma_\delta T_g)^2$ across the three gate times, so that the *collapse test* can check whether σ_ϕ alone controls the result. E7 runs at a detuning noise that gives $\sigma_\phi^2 = 0.04$, inside the single-channel equivalence region, so that any divergence must come from the second channel or from the correlation. At $\sigma_\gamma = 0.127$ the amplitude channel has the same phase variance as the detuning channel. E8 is described in Sec. IV E and E9 in Sec. IV B.

ID	Name	What is varied	Points
E1	Audit	$\alpha \in \{0.01, 0.025, 0.05, 0.1, 0.2\}$ at $T_g = 12.7$ ns, 4 levels	1
E2	Leakage sweep	$T_g = 6.3$–32.5 ns across the leakage onset; 4 and 2 levels	9+5
E3	Channel split	detuning only vs amplitude only, $T_g = 12.7$–20.8 ns	3+3
E4	Phase-variance sweep	$\sigma_\delta/2\pi = 0.5$–25 MHz at $T_g = 8$, 12.7, 20 ns; 2 levels, detuning only	16
E5	Strong search	20 restarts, 1000 iterations, $T_g = 15.4$, 20.8, 32.5 ns; 4 and 2 levels	3+3
E6	Replication	E1 and two E4 points with two further noise seeds	6
E7	Several channels	$\sigma_\gamma = 0.01$–0.127 and $\rho = -0.99$ to 0.99 at $T_g = 12.7$ ns, 2 levels; replication seeds	16
E8	Prospective test	four new points at $\rho = 0.99$, predictions fixed in advance	4
E9	Dominance test	candidate pools of 376–491 controls; E4's phase variances at 8, 12.7, 20 ns (2 levels, 3 seeds at 12.7 ns) and 12.7, 20.8 ns (4 levels)	42

TABLE S5. Experiments. Every point comprises one mean optimization and CVaR optimizations at $\alpha = 0.05$ and 0.2 (all five levels in E1), each with its own restarts and reverse start, all re-evaluated in the full Lindblad model.

Computation. All simulations were run on a personal laptop (10 CPU cores, 8 worker processes, Python with NumPy and SciPy). A two-level Stage-1 point (one mean and two CVaR multistarts with reverse starts) takes 70–90 s with the fast model, and a full-Lindblad re-evaluation of one point about 14 s.

S7. ADDITIONAL RESULTS

A. Leakage onset

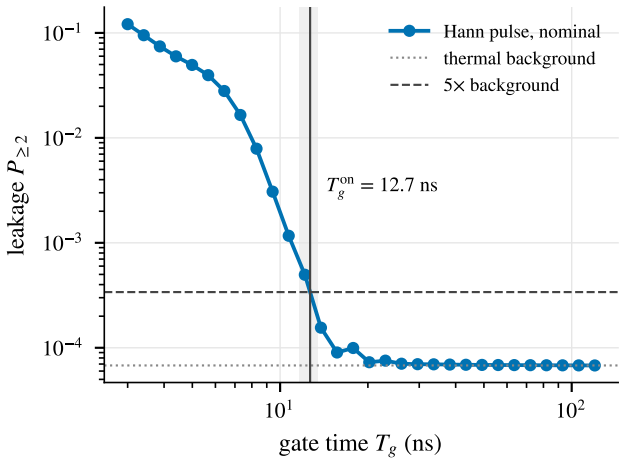


FIG. S3. Nominal leakage of a Hann pulse versus gate time. The onset (5× the thermal background of 6.8×10^{-5}) is at $T_g^{\mathrm{on}} = 12.7$ ns, and at 11.6 or 13.4 ns for 10× or 3×, so the onset does not hinge on the threshold chosen.

Leakage of a Hann pulse falls from 12% at 3 ns to the thermal floor by ∼20 ns (Fig. S3). The onset, $T_g^{\mathrm{on}} = 12.7$ ns, defines the leakage-dominated region $T_g < T_g^{\mathrm{on}}$.

T_g (ns)	6.3	7.4	8.5	9.9	11.5	12.7	15.4	20.8	32.5
Hann leakage	3×10^{-2}	2×10^{-2}	8×10^{-3}	3×10^{-3}	1×10^{-3}	5×10^{-4}	2×10^{-4}	7×10^{-5}	7×10^{-5}
R_{tail}, 4 levels	−0.003	−0.011	−0.016	−0.017	−0.021	−0.022	+0.104	+0.102	+0.064
R_{tail}, 2 levels	−0.026	–	–	–	–	−0.026	−0.012	−0.007†	+0.009
strong search, 4 lev.							+0.100	+0.117*	+0.059
strong search, 2 lev.							−0.008*	−0.021	−0.004

TABLE S6. E2 and E5, $\alpha = 0.05$, full Lindblad, CVaR control against the best mean control. Below and at the onset, $|R_{\text{tail}}| \leq 0.026$ in every case. Strong search: 20 restarts per objective and 1000 iterations. *A reverse start improved the mean control by more than 0.1%, and the value shown is against that improved control (two-level 15.4 ns: +0.076 at equal budget). 95% CIs of the strong search: $[0.077, 0.122]$, $[0.092, 0.138]$, $[0.036, 0.080]$. †Against the multistart mean control alone this cell reads +0.098; a reverse start found a mean control that matches the CVaR control's tail, so the apparent divergence was a missed basin (pitfall 6, Sec. S3.D).

B. Gate-time sweep and strong search

Throughout the leakage-dominated region, $6.3 \leq T_g \leq 12.7$ ns, the mean and CVaR controls coincide within $|R_{\text{tail}}| \leq 0.026$, with 4 levels and with 2 (Table S6). Where R_{tail} differs from zero at all it is slightly negative, the same estimation-variance penalty seen in the audit. The CVaR optimizer started from u_{mean} moves the pulse by only 1–2% and gains under 1% in-sample. This is not because leakage is absent from the problem: a Hann pulse at 6.3 ns leaks 3%. Both optimizers suppress leakage to near the thermal floor (≈ 7–9×10^{-5}), and the residual loss distributions remain ordered, exactly as Corollary 1 requires.

C. Channel split

T_g (ns)	12.7	15.4	20.8
both channels	−0.022	+0.104	+0.102
amplitude only	−0.013	−0.166‡	−0.001
detuning only	−0.015	+0.074	+0.177

TABLE S7. E3, 4 levels, $\alpha = 0.05$, full Lindblad, against the best mean control. Amplitude noise alone never produces a positive divergence. ‡Comparator asymmetry: the best mean control here came from a reverse start off the $\alpha = 0.2$ CVaR control, a basin the $\alpha = 0.05$ CVaR search never visited; at equal budget the cell reads −0.007.

Amplitude noise alone gives no CVaR advantage at any gate time (Table S7); its one large negative cell is a comparator artefact, not a physical effect. This rules out an alternative mechanism in which the level-2 AC Stark shift converts amplitude error into a quadratic phase. Any divergence requires detuning noise.

The strong search (E5) settles whether the long-gate divergence is real. With 20 restarts and 1000 iterations per objective, and CVaR compared against the best mean control found anywhere, the divergence persists: $R_{\text{tail}} = 0.100$, 0.117 and 0.059 at 15.4, 20.8 and 32.5 ns (Table S6). At 15.4 and 32.5 ns the reverse start returns to u_{mean}, so the CVaR solution is a genuine shift within the same basin.

a. The transmon diverges earlier than the qubit. A two-level qubit with exactly the same noise shows no divergence at any gate time up to 32.5 ns. With the same 20-restart, 1000-iteration search as the transmon, $R_{\text{tail}} = -0.008$, -0.021 and -0.004 at 15.4, 20.8 and 32.5 ns ($\sigma_\phi^2 = 0.06$, 0.11, 0.26), where the transmon gives 0.10, 0.12 and 0.06. The 4-restart runs agree (-0.012, -0.007, $+0.009$). At 15.4 ns the qubit shows once more how an apparent divergence arises: at equal budget $R_{\text{tail}} = +0.076$, but a reverse start finds a mean control with the same tail. In absolute terms, at 20.8 ns the transmon's tail gap is $D = 1.2 \times 10^{-3}$, against -4×10^{-5} for the qubit. The higher transmon levels therefore lower the divergence threshold and amplify it once detuning noise is present, while by themselves (amplitude noise only) they produce none.

$\sigma_\delta/2\pi$ (MHz)	0.5	1	2.5	5	8	16
σ_ϕ^2	0.0016	0.0064	0.040	0.16	0.41	1.6
$R_{\text{tail}}(0.05)$	$-0.001^\dagger$	-0.002	$+0.002$	$+0.029$	$+0.077$	$+0.275$
95% CI		$[-0.005, 0.001]$	$[-0.003, 0.007]$	$[0.026, 0.032]$	$[0.068, 0.085]$	$[0.247, 0.298]$
$R_{\text{tail}}(0.20)$	$-0.026^\dagger$	$+0.000$	$+0.002$	$+0.013$	$+0.034$	$+0.135$

TABLE S8. E4, 2 levels, detuning noise only, $T_g = 12.7$ ns, full Lindblad. The reverse warm start returns to u_{mean} at 5, 8 and 16 MHz (same basin), so these are genuine in-basin shifts. †Losses $\sim 10^{-5}$, at the optimizer's precision floor; the comparator was upgraded by a reverse start.

D. Phase-variance sweep and replication

On a bare qubit with detuning noise alone, the setting closest to Theorem 1, R_{tail} is zero within its confidence interval for $\sigma_\phi^2 \leq 0.04$. This is direct numerical confirmation of the theorem in its stated regime (Sec. S1). Beyond $\sigma_\phi^2 \approx 0.1$ it grows smoothly and monotonically, reaching 0.275 at $\sigma_\phi^2 = 1.6$ (Table S8). At every divergent point the mean optimizer, restarted from u_{CVaR}, walks back to u_{mean}: the two optima sit in one basin, and CVaR genuinely shifts the solution within it.

a. Replication. Repeating two points with two new noise seeds (E6): at $\sigma_\phi^2 = 1.6$, $R_{\text{tail}}(0.05) = 0.275$, 0.271 and 0.314 for the three seeds, all with confidence intervals far from zero. At $\sigma_\phi^2 = 0.16$, $R_{\text{tail}}(0.05) = 0.029$ and 0.036 for two seeds. For the third, the multistart comparison gives $+0.050$, but a reverse start from the $\alpha = 0.2$ CVaR control found a mean control with a better $\alpha = 0.05$ tail than the $\alpha = 0.05$ CVaR search reached ($R_{\text{tail}} = -0.40$ against it): a CVaR search failure of the kind described in pitfall 7, not a reversal of the effect.

E. Several channels

Amplitude width, $\rho = 0$						
σ_γ	0.01	0.02	0.04	0.08	0.127	0.127, no detuning
$R_{\text{tail}}(0.05)$	-0.134°	-0.029	-0.013	$+0.044$	$+0.017$	$+0.084^\circ$
95% CI	$[-0.16, -0.11]$	$[-0.06, 0.00]$	$[-0.03, 0.01]$	$[0.01, 0.08]$	$[-0.01, 0.04]$	$[-0.04, 0.18]$
κ	0.08	0.10	0.12	0.23	0.53	–
Correlation, $\sigma_\gamma = 0.127$						
ρ	-0.99	-0.9	-0.5	$+0.5$	$+0.9$	$+0.99$
$R_{\text{tail}}(0.05)$	$+0.226$	$+0.167$	$+0.114$	$+0.013$	$+0.077$	$+0.289$
95% CI	$[0.13, 0.30]$	$[0.08, 0.24]$	$[0.05, 0.17]$	$[-0.01, 0.04]$	$[0.07, 0.09]$	$[0.24, 0.34]$
κ	1.98	1.12	0.66	0.47	0.83	1.73

TABLE S9. E7, 2 levels, $T_g = 12.7$ ns, full Lindblad, against the best mean control. $\kappa = |Q|/|A|$ for the mean control. Replication seeds: at $\sigma_\gamma = 0.08$, $+0.028$ $[0.01, 0.05]$ and $+0.003$ $[-0.02, 0.02]$; at $\rho = +0.99$, $+0.073$ $[-0.08, 0.20]$ and $+0.334$ $[0.29, 0.37]$. $^\circ$Comparator upgraded by a reverse start.

a. P1 and P2: the two-axis mechanism is real but weak. With both channels independent, the objectives stay within a few percent of each other over the whole amplitude range (Table S9). The one significant positive point, $\sigma_\gamma = 0.08$, replicates weakly ($+0.044$, $+0.028$, $+0.003$ over three seeds). The geometric signature Proposition 2 predicts, a more isotropic CVaR control, appears in the first seed (loss anisotropy $A = 0.87 \to 0.79$) but not in the other two ($0.75 \to 0.74$, $0.88 \to 0.89$). The two-axis mechanism is therefore allowed by the theory but contributes little in practice: optimized controls can change the split of their error between the axes only slightly. Amplitude noise alone at $\sigma_\gamma = 0.127$ gives no significant divergence either.

b. P3's condition fails, and the reason is informative. Read unconditionally, as it was first framed, P3 fails: correlating the channels does not restore equivalence but produces the largest divergences of the whole study, $R_{\text{tail}} = 0.23$ at $\rho = -0.99$ and 0.29 at $\rho = +0.99$, with every reverse start returning to u_{mean}. The error clouds show why. With correlated noise the pulse need only cancel one combined first-order error direction instead of two, and it does: the first-order response along the combined noise direction $x_1 \pm x_2$ (the one that dominates as $|\rho| \to 1$) falls from 0.076 at $\rho = 0$ to 0.013–0.017 at $|\rho| = 0.99$, and the mean loss of u_{mean} drops fourfold. What remains is second order

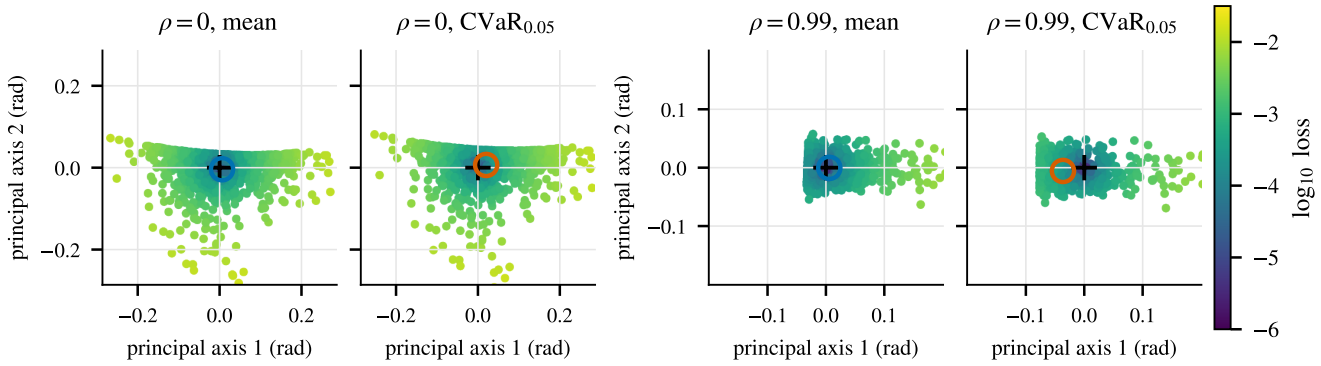


FIG. S4. Error clouds: the error rotation vector of each of the 1000 test draws, projected onto its two principal axes and coloured by loss, for the mean and $\mathrm{CVaR}_{0.05}$ controls at $\sigma_\gamma = 0.127$. Cross: zero error. Circle: the average error of the cloud. Left pair ($\rho = 0$): the cloud is two-dimensional, curved by second-order error, and the two controls are almost identical. Right pair ($\rho = 0.99$): the linear spread is cancelled along one direction and the cloud collapses to a thin, one-sided streak. The CVaR control shifts the whole streak (circle to the left of the cross), so that its long right-hand tail is pulled back towards zero at the cost of the typical draw.

in the single noise variable, so the linear-coupling condition of P3 fails, with κ above 1. Figure S4 shows the difference directly: the independent-channel cloud is a broad two-dimensional fan, whereas the correlated-channel cloud is a narrow, one-sided streak whose extent is set by the second-order term. Rank-one *noise* is not rank-one *linear error*: the optimization itself removes the linearity that equivalence needs. One of three seeds at $\rho = +0.99$ is not significant ($+0.073$, with a wide interval); the other two are.

c. How different are the pulses? Divergent CVaR controls are small modifications of the mean controls (Fig. S5). At the strongest Stage-1 divergence the relative change in the control envelopes is 6.6%, against 0.8% at a non-divergent point; in the gate-time sweep (E2) the CVaR optimizer moved the pulse by 1–2% in the equivalence region. Wherever the reverse start returns to u_{mean}, the two objectives select nearby points in one basin, not different pulse families. The effect of the small change on the error is nonetheless large and systematic, as the next section shows.

F. Independence of the experimental parameters

covariate	marginal	given κ	given R^{model}
$\lvert\rho\rvert$	$+0.70$	-0.00 ($p = 0.98$)	$+0.08$ ($p = 0.69$)
detuning phase variance σ_ϕ^2	$+0.09$	$+0.24$ ($p = 0.22$)	-0.03 ($p = 0.90$)
amplitude phase variance	$+0.53$	-0.19 ($p = 0.32$)	$+0.28$ ($p = 0.15$)
total phase variance	$+0.41$	$+0.25$ ($p = 0.20$)	$+0.10$ ($p = 0.60$)
gate time T_g	-0.29	-0.38 ($p = 0.05$)	-0.15 ($p = 0.45$)

TABLE S10. Does κ carry the information of the experimental knobs? Spearman correlation of $R_{\mathrm{tail}}(0.05)$ with each covariate, marginal and partial (controlling for κ or R^{model}), 28 two-level points, including the prospective test. Conversely, κ keeps a partial correlation of 0.61–0.84 ($p \le 6 \times 10^{-4}$) and R^{model} of 0.78–0.89 ($p \le 10^{-6}$) against every covariate.

a. Independence of the knobs. Correlation $|\rho|$ predicts R_{tail} on its own (Spearman 0.70), but once κ is known it adds nothing (partial -0.00, $p = 0.98$); the same holds for noise strength (Table S10). Gate time keeps a borderline residual given κ (-0.38, $p = 0.05$, one of ten tests), driven by the 8 ns point, which diverges strongly at a moderate $\kappa = 0.82$. It disappears given R^{model} ($p = 0.45$): the extra information is in the error geometry, which κ discards and the reduced model keeps. Within these data, ρ, T_g and noise strength affect the divergence only through κ and the error geometry, as Proposition 5 requires. This test is observational: the covariates were not varied independently of one another, and the two-level data cover T_g only between 6.3 and 32.5 ns.

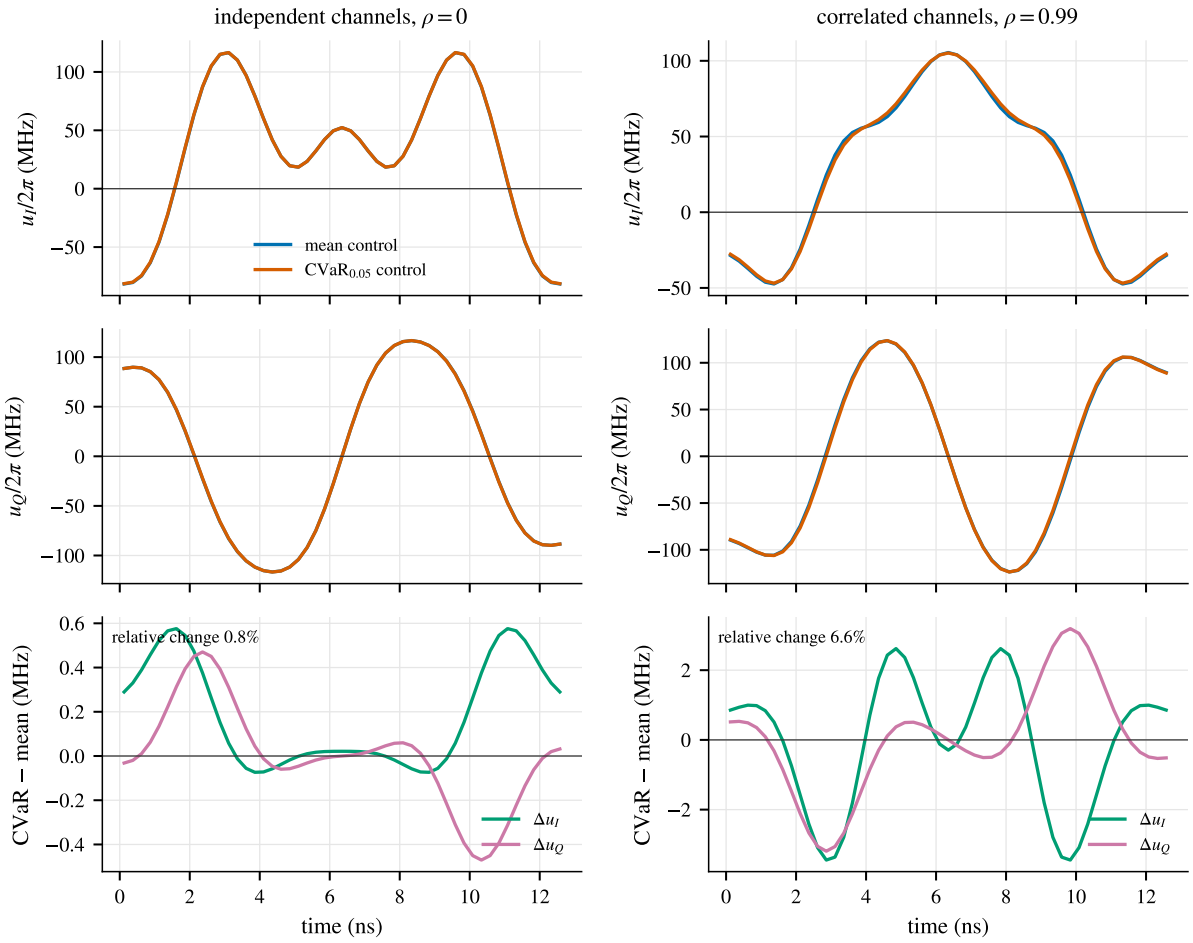


FIG. S5. Mean and CVaR$_{0.05}$ controls at $\sigma_\gamma = 0.127$ (filtered envelopes as applied to the qubit). Top two rows: the in-phase and quadrature envelopes, visually indistinguishable at this scale. Bottom row: their difference. With independent channels (left) the CVaR control differs by 0.8% in norm; with correlated channels (right) by 6.6%, with an oscillating correction of a few MHz, the change that produces a 29% lower tail infidelity.

G. Pre-compensation offsets

The offsets of Fig. 9(c), per point (coherent offset $c = -\mathbf{e}_0 \cdot \mathbf{q}/|\mathbf{q}|^2$ of each control, in units of its second-order error): detuning scan at 5, 8 and 16 MHz: mean control 0.51, 0.51, 0.49; CVaR$_{0.2}$ 1.61, 1.62, 1.71; CVaR$_{0.05}$ 2.46, 2.48, 2.54. Correlated channels, CVaR$_{0.05}$: 2.00 ($\rho = -0.99$), 1.87 (-0.9), 1.02 ($+0.9$), 1.98 and 2.52 ($+0.99$, two seeds), and 1.63, 1.94, 2.03, 2.46 at the four points of the prospective test; CVaR$_{0.2}$ 1.15–1.80; mean controls 0.73–0.94. The exception $\rho = +0.9$ retains a large second noise direction.

S8. DOMINANCE TEST: DETAILS

Pools and statistics. The candidate pool of a setting contains a geometrically thinned subset of the L-BFGS-B iterates of every run (mean, CVaR at $\alpha = 0.05$ and 0.2, reverse starts) plus every run's final control, deduplicated. Every candidate is evaluated with the closed-system simulator on a fresh ensemble of 4000 draws (seed $44 + 1000s$), split into two halves of 2000. Statistics are computed over the competitive candidates (mean loss within a factor of two of the pool's best). The full-Lindblad R_{tail} of the standard comparison is computed as in the rest of the study. A confirmed reversal requires a separation of more than 2% in both mean and CVaR$_{0.05}$ on both halves, with the same orientation on both halves. The pool regret selects the best-mean and best-CVaR candidates on one half, scores both by CVaR$_{0.05}$ on the other, $R_{\text{pool}} = 1 - \text{CVaR(best-CVaR)}/\text{CVaR(best-mean)}$, and averages the two assignments.

Transmon and second basin. With four levels the same pattern holds, shifted. Reversals stay near 1% at the smallest noise instead of zero, and the shape ratio is 1.8–3.1 instead of 5.58, because a noise-independent leakage floor makes up 56–84% of the mean loss there. At 20.8 ns the transmon crosses over earlier than the qubit (4.6% reversals at $\sigma_\phi^2 = 0.04$, 22% at 0.16). At 8 and 20 ns and $\sigma_\phi^2 = 1.6$ the competitive pool is dominated by a second basin, found by the reverse starts, with a four to five times lower mean loss but a heavier relative tail ($\mathrm{CVaR}_{0.05}/\mathrm{mean} \approx 10$).

The unrestricted crossing test. Table S11 lists, for every two-level setting with $\sigma_\phi^2 \le 0.04$, the crossing statistic as first specified (any crossing of the quantile functions by more than 2%, no mean separation required), together with the location of the crossings and the fraction of crossing pairs in which the control with the lower mean has the larger noise-free loss L_0. The crossings are all located in the lower half of the distribution, and in all but one seed they are driven by the coherent error. The lowest quantiles of a χ_1^2 loss are tiny (0.4% of the mean at $q = 0.05$), so a residual coherent error of a few tenths of a percent of the mean decides them, whereas the upper half, which determines both the mean and the CVaR, is set by the noise. The restricted statistic of the main text (pairs with at least 10% mean separation, crossings at $q \ge 0.5$) is zero at all of these settings.

T_g (ns)	seed	σ_ϕ^2	crossing pairs	fraction	median q	max q	lower mean has larger L_0
8	0	0.0016	250	0.113	0.06	0.19	0.96
8	0	0.0064	988	0.233	0.07	0.25	0.91
8	0	0.04	1343	0.268	0.19	0.45	0.91
12.7	0	0.0016	257	0.133	0.07	0.42	0.98
12.7	1	0.0016	141	0.192	0.10	0.36	0.96
12.7	2	0.0016	45	0.045	0.05	0.09	0.49
12.7	0	0.0064	194	0.119	0.08	0.27	0.84
12.7	1	0.0064	277	0.210	0.07	0.34	0.91
12.7	2	0.0064	107	0.103	0.07	0.12	0.29
12.7	0	0.04	663	0.215	0.17	0.41	0.96
12.7	1	0.04	1232	0.308	0.15	0.46	0.93
12.7	2	0.04	677	0.240	0.12	0.43	0.78
20	0	0.0016	64	0.062	0.08	0.16	0.97
20	0	0.0064	151	0.146	0.08	0.27	0.95
20	0	0.04	469	0.270	0.15	0.44	0.93

TABLE S11. Unrestricted crossing test at small noise, two levels. "median q" and "max q": median and maximum over crossing pairs of the quantile level of the largest crossing.

family	τ	reversals	R_{pool}
ordered, $\sin^2(\sigma_i Z/2)$	1	0	0
pre-compensation, $(Z^2 - c_i)^2$	−0.46	0.76	0.42

TABLE S12. Synthetic validation of the statistics (`stage2.py selftest`), run before the production run.

S9. THE CONTROLLED EXPERIMENT, LEAKAGE-AWARE κ, AND GENERALITY

A. Design and calibration

κ is a property of the mean-optimal control and is therefore known only after an optimization, so a design that holds it fixed has to solve for the noise level that produces it. For each cell $(\kappa^\star, \rho, T_g)$ of the grid $\kappa^\star \in \{0.20, 0.35, 0.55\}$, $\rho \in \{0, 0.5, 0.9\}$, $T_g \in \{8, 12.7, 20\}$ ns we scale the base widths $(\sigma_\delta, \sigma_\gamma) = (2.5\ \mathrm{MHz}, 0.127)$ by one number s and solve $\kappa(s) = \kappa^\star$ by secant iteration in $\log s$, seeded with the empirical scaling $\kappa \propto s^{1/2}$ and stopped at $|\log \kappa/\kappa^\star| < \log 1.05$ or six evaluations. Each evaluation is a complete mean multistart plus a fit of Eq. (10) on the test ensemble. The 27 cells took 112 mean optimizations; 21 converged inside the 5% tolerance, and the worst cell ended 25% away.

Each calibrated cell is then run with the standard protocol of Sec. S3 at two noise seeds, and the calibrated control is offered to the mean side as one of its restarts (the CVaR side is warm-started from the selected mean control as always, so this keeps the comparison conservative). The run's mean control is nevertheless better optimized than the calibration's and cancels more linear error, so its κ is larger: by a median factor 1.08 over the 54 points, 1.30 at $T_g = 8$ ns, and up to 1.74 in one cell. The achieved ranges are 0.18–0.33, 0.35–0.56 and 0.51–0.74 for the three rows, which overlap slightly; all analyses use the measured κ of the control that was run.

Statistics. Because a cell can return $R_{\mathrm{tail}} \le 0$, ratios to a prediction are not usable (a clipped ratio turns "no gain" into an arbitrarily large log-residual), and with three cells per knob value a Kruskal–Wallis test has little power. The tests reported in the main text are therefore partial rank correlations of R_{tail} with a knob, controlling for κ or for R^{model}, computed over all 54 points and over the 36 whose mean loss exceeds the T_1/T_2 floor of 3×10^{-4}. Below that floor the CVaR estimate is dominated by its six training draws: six of the eighteen $\rho = 0.9$ cells return a significantly negative R_{tail}.

B. Leakage-aware error coordinates

Let U be the full N-level propagator of a draw and $E = U_{\mathrm{target}}^{\dagger} U$ with the target embedded as the identity on levels ≥ 2. Write E_q for the qubit block and $l_{kj} = E_{kj} = U_{kj}$, $k \ge 2$, $j \in \{0, 1\}$ for the leakage amplitudes. Taking the polar factor W_u of E_q, normalizing it to $SU(2)$ and applying the *same* global phase to l (a gauge choice, since the loss is invariant under it) gives the coordinate $x = (\boldsymbol{\epsilon}; \mathrm{Re}\, l; \mathrm{Im}\, l) \in \mathbb{R}^{3+4(N-2)}$.

Since U is unitary, $E_q^{\dagger} E_q + l^{\dagger} l = I$, so $E_q = W_u (I - l^{\dagger} l)^{1/2} \approx W_u (I - \frac{1}{2} l^{\dagger} l)$ and, for an input ψ in the computational subspace,

$$F_{\psi} = |\langle \psi | E_q | \psi \rangle|^2 \approx 1 - \tfrac{1}{4} \left[\epsilon^2 - (\boldsymbol{\epsilon} \cdot \mathbf{r})^2 \right] - \langle \psi | l^{\dagger} l | \psi \rangle .$$

Averaging over the four inputs gives the metric of Sec. S2.E for the rotation part and $\bar{P}_{\mathrm{leak}} = \frac{1}{4} \sum_{\psi} \langle \psi | l^{\dagger} l | \psi \rangle$ for the leakage part; the loss $L = 1 - \bar{F} + \lambda \bar{P}_{\mathrm{leak}}$ therefore carries $(1 + \lambda)$ on the leakage term. With $S = \sum_{\psi} |\psi\rangle\langle\psi| = 2I + \frac{1}{2}(X + Y)$ and $G = \bar{S}$,

$$\bar{P}_{\mathrm{leak}} = \tfrac{1}{4} \sum_{k \ge 2} l_k^{\dagger} G \, l_k ,$$

which in the real coordinates $(\mathrm{Re}\, l, \mathrm{Im}\, l)$ is the positive-definite form quoted in the main text. `stage3.py selftest` checks it against the exact four-state loss of random unitaries: the relative error is 2×10^{-5} at an error size of 3×10^{-4} and falls as the cube of that size, as a second-order metric must.

The mechanism variable in these coordinates is $\kappa_{\mathrm{lk}} = \|W^{1/2} Q\|_F / \|W^{1/2} A\|_F$, and the reduced model is evaluated in the same space, with the offset free in all coordinates. For $N = 2$ the leakage block is empty and κ_{lk} reduces to the metric-weighted κ_W, which differs from κ only by the weighting $W_{\epsilon} = \mathrm{diag}(3, 3, 2)/16$.

C. Generality grid

The target rotation is installed by rebuilding the closed-system model with $U_{\mathrm{target}} = \exp(-i\theta\, \hat{n} \cdot \boldsymbol{\sigma}/2)$ embedded in the N-level space; the amplitude budget scales with θ, so X_{π} is as tightly constrained as $X_{\pi/2}$, and the Hann seed is scaled to the correct area. Bases are (i) the M Hann-windowed sine modes per quadrature of Sec. S2, (ii) $M = 6$ of them, and (iii) M windowed piecewise-constant segments, which matches the parameter count of (i) at $M = 17$. Anharmonicity is varied on the four-level model only.

Two caveats limit what these points can be compared with. First, the Lindblad worker has the $X_{\pi/2}$ target at -200 MHz compiled into it, so points with another target or anharmonicity are closed-system only; their JSON records this and the validation step skips them. Second, and consequently, the best-mean comparator of Sec. S3, which the Lindblad validation applies, had to be applied separately to the saved closed-system losses (`stage3_recomp.py`). It changed 46 of the 76 Stage-3 points, by more than 0.05 at four of them; at the two-level $X_{\pi/2}$ point with 17 modes it removed a spurious $R_{\mathrm{tail}} = +0.31$, bringing it to -0.02 in agreement with the phase-variance sweep at the same setting. The numbers reported in the main text and in Table S14 are the corrected ones.

TABLE S13. The controlled experiment, all 54 points. κ_{cal}: the calibration control; κ_{run}: the control actually run; C: the coefficient of Eq. (13); R^{model}: reduced model; R_{tail}: closed-system, best-mean comparator, $\alpha = 0.05$. σ_δ in MHz.

$\kappa^\star$	ρ	T_g	seed	σ_δ	σ_γ	κ_{cal}	κ_{run}	C	R^{model}	R_{tail}	$\mathbb{E}L$
0.20	0.0	8	0	2.41	0.122	0.257	0.310	1.98	0.123	+0.094	2e-03
0.20	0.0	8	1	2.41	0.122	0.257	0.280	3.57	0.164	+0.138	2e-03
0.20	0.0	12.7	0	1.26	0.064	0.206	0.245	0.22	0.018	−0.025	4e-04
0.20	0.0	12.7	1	1.26	0.064	0.206	0.236	0.13	0.011	−0.055	4e-04
0.20	0.0	20	0	1.28	0.065	0.193	0.194	0.74	0.023	−0.005	8e-04
0.20	0.0	20	1	1.28	0.065	0.193	0.178	1.25	0.032	+0.012	8e-04
0.20	0.5	8	0	1.54	0.078	0.209	0.274	1.32	0.087	−0.005	5e-04
0.20	0.5	8	1	1.54	0.078	0.209	0.271	1.49	0.091	+0.050	6e-04
0.20	0.5	12.7	0	0.83	0.042	0.207	0.228	0.07	0.007	−0.050	1e-04
0.20	0.5	12.7	1	0.83	0.042	0.207	0.224	0.02	0.004	−0.016	1e-04
0.20	0.5	20	0	0.91	0.046	0.191	0.206	0.12	0.006	−0.003	2e-04
0.20	0.5	20	1	0.91	0.046	0.191	0.216	0.06	0.003	+0.025	2e-04
0.20	0.9	8	0	1.04	0.053	0.188	0.327	0.24	0.071	−0.050	8e-05
0.20	0.9	8	1	1.04	0.053	0.188	0.327	0.25	0.072	−0.021	8e-05
0.20	0.9	12.7	0	0.41	0.021	0.208	0.217	0.01	0.005	−0.030	7e-06
0.20	0.9	12.7	1	0.41	0.021	0.208	0.205	0.00	0.003	−0.160	8e-06
0.20	0.9	20	0	0.49	0.025	0.209	0.227	0.01	0.007	+0.001	2e-05
0.20	0.9	20	1	0.49	0.025	0.209	0.224	0.00	0.005	+0.001	2e-05
0.35	0.0	8	0	3.24	0.165	0.365	0.394	3.19	0.219	+0.176	4e-03
0.35	0.0	8	1	3.24	0.165	0.365	0.390	4.48	0.211	+0.187	4e-03
0.35	0.0	12.7	0	1.97	0.100	0.357	0.376	0.12	0.036	−0.027	1e-03
0.35	0.0	12.7	1	1.97	0.100	0.357	0.371	0.14	0.040	−0.100	1e-03
0.35	0.0	20	0	2.67	0.136	0.428	0.367	0.07	0.012	−0.008	4e-03
0.35	0.0	20	1	2.67	0.136	0.428	0.401	0.47	0.059	+0.036	4e-03
0.35	0.5	8	0	2.21	0.112	0.355	0.379	2.15	0.171	+0.147	1e-03
0.35	0.5	8	1	2.21	0.112	0.355	0.373	1.53	0.150	+0.082	1e-03
0.35	0.5	12.7	0	1.52	0.077	0.357	0.385	0.02	0.034	+0.035	4e-04
0.35	0.5	12.7	1	1.52	0.077	0.357	0.415	0.03	0.050	+0.053	5e-04
0.35	0.5	20	0	1.58	0.080	0.341	0.351	0.08	0.022	−0.016	8e-04
0.35	0.5	20	1	1.58	0.080	0.341	0.352	0.18	0.046	−0.026	9e-04
0.35	0.9	8	0	1.91	0.097	0.345	0.561	2.94	0.326	+0.226	4e-04
0.35	0.9	8	1	1.91	0.097	0.345	0.555	3.12	0.331	+0.039	4e-04
0.35	0.9	12.7	0	0.70	0.035	0.347	0.380	0.00	0.046	−0.022	2e-05
0.35	0.9	12.7	1	0.70	0.035	0.347	0.383	0.00	0.047	−0.049	2e-05
0.35	0.9	20	0	0.77	0.039	0.359	0.370	0.04	0.053	−0.082	4e-05
0.35	0.9	20	1	0.77	0.039	0.359	0.371	0.00	0.037	+0.011	5e-05
0.55	0.0	8	0	6.37	0.324	0.503	0.742	3.40	0.327	+0.190	2e-02
0.55	0.0	8	1	6.37	0.324	0.503	0.744	3.69	0.317	+0.183	2e-02
0.55	0.0	12.7	0	3.20	0.162	0.556	0.610	0.13	0.124	−0.036	4e-03
0.55	0.0	12.7	1	3.20	0.162	0.556	0.559	0.18	0.115	+0.101	4e-03
0.55	0.0	20	0	3.74	0.190	0.530	0.521	0.55	0.084	+0.030	9e-03
0.55	0.0	20	1	3.74	0.190	0.530	0.508	1.59	0.169	+0.120	9e-03
0.55	0.5	8	0	4.03	0.205	0.465	0.656	3.25	0.306	+0.202	5e-03
0.55	0.5	8	1	4.03	0.205	0.465	0.716	1.74	0.294	+0.068	5e-03
0.55	0.5	12.7	0	2.08	0.106	0.533	0.530	0.02	0.093	+0.039	9e-04
0.55	0.5	12.7	1	2.08	0.106	0.533	0.574	0.02	0.099	+0.023	9e-04
0.55	0.5	20	0	2.55	0.129	0.529	0.528	0.03	0.055	+0.007	3e-03
0.55	0.5	20	1	2.55	0.129	0.529	0.578	0.06	0.083	+0.074	3e-03
0.55	0.9	8	0	2.14	0.109	0.488	0.600	4.87	0.356	+0.264	5e-04
0.55	0.9	8	1	2.14	0.109	0.488	0.613	2.53	0.337	+0.086	6e-04
0.55	0.9	12.7	0	1.07	0.054	0.560	0.593	0.01	0.151	+0.015	6e-05
0.55	0.9	12.7	1	1.07	0.054	0.560	0.577	0.00	0.152	−0.132	6e-05
0.55	0.9	20	0	1.35	0.068	0.576	0.628	0.12	0.197	+0.041	2e-04
0.55	0.9	20	1	1.35	0.068	0.576	0.609	0.00	0.150	−0.011	2e-04

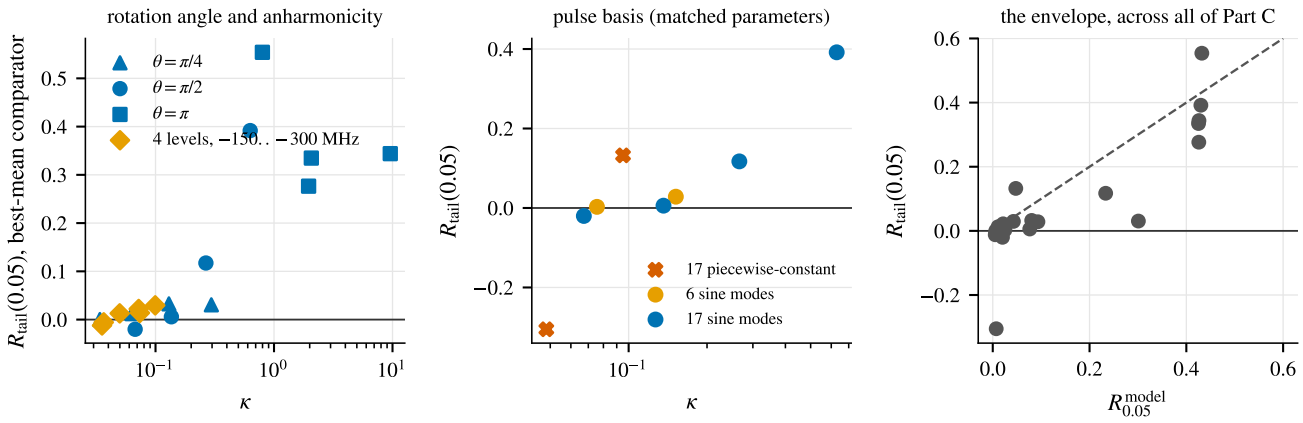


FIG. S6. The generality grid. (a) R_{tail} against κ for the three rotation angles (two-level, 17 sine modes) and for the four-level anharmonicity scan. (b) The pulse-basis comparison at $X_{\pi/2}$: 17 sine modes against 17 piecewise-constant segments (the same number of parameters) and 6 sine modes. The two piecewise-constant points are the outliers, one of them at $R_{\mathrm{tail}} = -0.31$. (c) Measured against the reduced model over all 22 points.

TABLE S14. The generality grid, all 22 points; anharmonicity in MHz, basis as described above, noise scale relative to (2.5 MHz, 0) at $T_g = 12.7$ ns. Closed-system values with the best-mean comparator.

levels	θ	anh.	basis	scale	κ	C	R^{model}	R_{tail}	$\mathbb{E}L$
2	π	-200	17 sine	1	9.573	7.88	0.426	+0.344	1e-07
2	π	-200	17 sine	2	0.798	8.15	0.432	+0.554	1e-08
2	π	-200	17 sine	4	1.958	9.43	0.426	+0.277	8e-07
2	π	-200	17 sine	8	2.065	10.17	0.425	+0.335	2e-05
2	$\pi/2$	-200	17 p.c.	1	0.048	2.82	0.007	−0.305	1e-04
2	$\pi/2$	-200	17 p.c.	2	0.095	5.37	0.048	+0.132	3e-04
2	$\pi/2$	-200	6 sine	1	0.076	4.52	0.025	+0.003	2e-04
2	$\pi/2$	-200	6 sine	2	0.152	4.53	0.093	+0.028	9e-04
2	$\pi/2$	-200	17 sine	1	0.067	4.53	0.020	−0.020	3e-04
2	$\pi/2$	-200	17 sine	2	0.136	4.53	0.076	+0.006	1e-03
2	$\pi/2$	-200	17 sine	4	0.266	4.53	0.233	+0.117	5e-03
2	$\pi/2$	-200	17 sine	8	0.630	4.53	0.430	+0.392	1e-02
2	$\pi/4$	-200	17 sine	1	0.034	5.26	0.006	+0.000	3e-03
2	$\pi/4$	-200	17 sine	2	0.063	5.28	0.021	+0.012	1e-02
2	$\pi/4$	-200	17 sine	4	0.130	5.28	0.080	+0.033	5e-02
2	$\pi/4$	-200	17 sine	8	0.296	5.31	0.301	+0.031	1e-01
4	$\pi/2$	-150	17 sine	1	0.037	4.59	0.006	−0.006	5e-03
4	$\pi/2$	-200	17 sine	1	0.035	4.51	0.005	−0.012	3e-03
4	$\pi/2$	-300	17 sine	1	0.050	4.63	0.011	+0.013	2e-03
4	$\pi/2$	-150	17 sine	2	0.074	4.60	0.024	+0.014	2e-02
4	$\pi/2$	-200	17 sine	2	0.072	4.50	0.021	+0.022	1e-02
4	$\pi/2$	-300	17 sine	2	0.099	4.66	0.043	+0.030	5e-03

S10. COLORED NOISE

A. Ornstein–Uhlenbeck sampling and normalization

The detuning is replaced by an Ornstein–Uhlenbeck process $d\eta = -\eta\, dt/\tau_c + \sigma_{\mathrm{ou}}\sqrt{2/\tau_c}\, dW$, with stationary variance σ_{ou}^2 and correlation $C(t,t') = \sigma_{\mathrm{ou}}^2 e^{-|t-t'|/\tau_c}$. On the control grid of n slices of width dt it is sampled exactly as a

stationary AR(1) chain,

$$\eta_{j+1} = a\eta_j + \sigma_{\rm ou}\sqrt{1-a^2}\,\xi_j, \qquad a = e^{-dt/\tau_c}, \tag{S25}$$

with $\eta_0 \sim \mathcal{N}(0, \sigma_{\rm ou}^2)$ and ξ_j i.i.d. standard normal, so no discretization error enters the noise itself. The accumulated phase over the gate is $\phi = dt\sum_j \eta_j$, whose variance is exactly

$$\mathrm{Var}(\phi) = \sigma_{\rm ou}^2\, dt^2 \sum_{j,k=0}^{n-1} a^{|j-k|} = \sigma_{\rm ou}^2\, dt^2\Big(n + 2\sum_{d=1}^{n-1}(n-d)a^d\Big). \tag{S26}$$

We solve this for $\sigma_{\rm ou}$ so that $\mathrm{Var}(\phi)$ equals the target σ_ϕ^2 at every τ_c. The quasi-static arm is the $\tau_c \to \infty$ limit of the same construction and reproduces the sampler of Sec. S2. Without this normalization a short-τ_c arm would simply be a weaker-noise arm, and the whole comparison would be confounded; the resulting $\sigma_{\rm ou}$ grows from 2.51 MHz quasi-static to 18.9 MHz at $\tau_c = 0.1\,T_g$, $\sigma_\phi^2 = 0.41$, as it must.

B. Noise coordinates

The error-geometry fit of Sec. S4 expands the error in whitened scalar noise coordinates. A trajectory has no single scalar, so we use the first two functionals of the path that a gate can resolve: the accumulated phase and its first time moment about the mid-gate instant,

$$z_1 \propto dt\sum_j \eta_j, \qquad z_2 \propto dt\sum_j (t_j - \bar{t})\,\eta_j, \tag{S27}$$

each centred over the ensemble, then z_2 orthogonalized against z_1 and both standardized. Under quasi-static noise z_2 vanishes identically, and the fit reduces to the scalar one. The reported $R^2_{\rm fit}$ is the smallest coefficient of determination over the components of the error vector when regressed on (z_1, z_2) and their quadratic products; it measures whether two coordinates suffice. They do not at short correlation time, which is the honest limitation of this stage: $R^2_{\rm fit}$ falls to 0.010 at $\tau_c = 0.1\,T_g$. A third and higher moments would extend the reduction, at the cost of a fit that needs many more draws to condition.

C. Design and scoring

The grid is $\tau_c/T_g \in \{0.1, 0.3, 1, 3, 10\}$ plus a quasi-static arm, crossed with $\sigma_\phi^2 \in \{0.04, 0.16, 0.41\}$ and seeds $\{0, 1\}$: 36 points, all at $T_g = 12.7$ ns on the two-level model, 17 sine modes, with the fairness protocol of Sec. S3 unchanged. The best-mean comparator is applied at save time rather than during validation, which is the fix for the pitfall recorded in Sec. S9.C; it upgraded 20 of the 36 points, by at most 0.064.

Four predictions were registered before the run. Two were met as stated (T1, memory independence; T4, the growth of the z_2 share), and two were mis-specified in ways the run exposed.

T2 was a special case. The registered prediction was $\mathrm{CVaR}_\alpha/\mathbb{E} = 5.58$ at every point. That is $\rho_\alpha(1)$, and holds only when one noise direction reaches the gate. Proposition 3 supersedes it; the quasi-static arm, where $m_{\rm eff} = 1$ by construction, returns 5.36 and does satisfy the original form.

T3 was unattainable. The registered threshold was $\mathrm{Spearman}(\kappa, R_{\rm tail}) > 0.6$. With two seeds per cell the replicate standard deviation of $R_{\rm tail}$ is $\sigma_{\rm noise} = 0.0205$ against a total 0.0256, leaving $\sigma_{\rm signal} = 0.0153$; classical attenuation then caps any observable correlation at $(1 + \sigma^2_{\rm noise}/\sigma^2_{\rm signal})^{-1/2} = 0.600$. The threshold demanded a perfect underlying correlation. We report disattenuated values instead: $\mathrm{Pearson}(R^{\rm model}, R_{\rm tail}) = 0.67$ observed, 1.12 disattenuated, and $\mathrm{Spearman}(\kappa, R_{\rm tail}) = 0.44$ observed, 0.74 disattenuated. The residual scatter about the fit is 0.0190, or 0.93 of the replicate scatter.

Table S15 lists every point. We record both re-specifications here rather than in the main text so that the original registration stays visible. Neither changes a conclusion in the direction that favours the theory: T2's correction makes the shape prediction harder, since it now must hold per control rather than as one number, and T3's makes the mechanism claim weaker, since it becomes a statement about consistency with a bound rather than about rank discrimination.

TABLE S15. All 36 colored-noise points, ordered by decreasing correlation time. $\tau_c/T_g = \infty$ is the quasi-static arm. C and R^{model} come from the two-coordinate reduction and should be read with R^2_{fit} in mind; ratio is $\text{CVaR}_{0.05}/\mathbb{E}$; the last column is the comparator selected by the best-mean rule of Sec. S3.A.

τ_c/T_g	σ_ϕ^2	seed	κ	C	$C\kappa^2$	R^{model}	R_{tail}	ratio	z_2 share	R^2_{fit}	comparator
10	0.04	0	0.065	4.92	0.0207	0.0204	+0.0089	4.40	0.270	0.984	mean
10	0.04	1	0.065	4.85	0.0204	0.0200	-0.0431	4.48	0.273	0.982	$\text{rev}_{0.2}$
10	0.16	0	0.130	4.93	0.0836	0.0738	+0.0278	4.36	0.287	0.983	mean
10	0.16	1	0.129	4.83	0.0800	0.0712	-0.0122	4.43	0.286	0.981	$\text{rev}_{0.05}$
10	0.41	0	0.208	4.83	0.2087	0.1557	+0.0599	4.32	0.305	0.980	$\text{rev}_{0.05}$
10	0.41	1	0.206	4.74	0.2001	0.1507	+0.0258	4.36	0.306	0.977	$\text{rev}_{0.2}$
3	0.04	0	0.053	1.83	0.0052	0.0053	-0.0092	3.98	0.551	0.961	mean
3	0.04	1	0.053	1.79	0.0051	0.0052	+0.0020	4.14	0.543	0.961	mean
3	0.16	0	0.105	1.69	0.0185	0.0193	-0.0092	3.96	0.555	0.944	mean
3	0.16	1	0.104	1.81	0.0196	0.0201	-0.0043	4.12	0.549	0.961	$\text{rev}_{0.2}$
3	0.41	0	0.165	1.28	0.0348	0.0413	-0.0028	3.95	0.586	0.962	$\text{rev}_{0.05}$
3	0.41	1	0.164	1.35	0.0362	0.0418	-0.0031	4.07	0.578	0.957	mean
1	0.04	0	0.041	0.33	0.0006	0.0007	-0.0372	4.21	0.703	0.853	$\text{rev}_{0.2}$
1	0.04	1	0.041	0.36	0.0006	0.0005	-0.0072	4.42	0.750	0.908	$\text{rev}_{0.05}$
1	0.16	0	0.077	0.13	0.0007	0.0013	+0.0000	4.33	0.771	0.909	mean
1	0.16	1	0.080	0.15	0.0009	0.0013	-0.0070	4.40	0.761	0.931	$\text{rev}_{0.05}$
1	0.41	0	0.121	0.04	0.0006	0.0024	-0.0027	4.29	0.790	0.883	mean
1	0.41	1	0.123	0.08	0.0012	0.0035	-0.0083	4.34	0.776	0.890	mean
0.3	0.04	0	0.022	2.63	0.0012	0.0013	-0.0032	3.65	0.602	0.390	$\text{rev}_{0.2}$
0.3	0.04	1	0.044	1.57	0.0030	0.0032	-0.0173	3.63	0.581	0.422	$\text{rev}_{0.2}$
0.3	0.16	0	0.044	1.90	0.0037	0.0037	+0.0027	3.63	0.604	0.389	mean
0.3	0.16	1	0.064	1.87	0.0077	0.0078	-0.0380	3.61	0.540	0.495	mean
0.3	0.41	0	0.075	1.47	0.0083	0.0080	+0.0077	3.58	0.607	0.395	mean
0.3	0.41	1	0.093	1.82	0.0159	0.0146	-0.0323	3.57	0.543	0.504	$\text{rev}_{0.2}$
0.1	0.04	0	0.018	3.23	0.0011	0.0013	-0.0050	3.83	0.652	0.196	$\text{rev}_{0.2}$
0.1	0.04	1	0.069	0.19	0.0009	0.0009	-0.0077	3.44	0.669	0.264	$\text{rev}_{0.05}$
0.1	0.16	0	0.039	0.76	0.0012	0.0013	-0.0147	3.65	0.789	0.062	$\text{rev}_{0.2}$
0.1	0.16	1	0.074	0.11	0.0006	0.0008	-0.0017	3.49	0.797	0.092	mean
0.1	0.41	0	0.057	0.34	0.0011	0.0013	-0.0077	3.77	0.844	0.010	$\text{rev}_{0.2}$
0.1	0.41	1	0.112	0.36	0.0046	0.0042	+0.0174	3.33	0.704	0.213	$\text{rev}_{0.2}$
∞	0.04	0	0.074	4.53	0.0247	0.0240	+0.0040	5.17	0.000	1.000	mean
∞	0.04	1	0.075	4.53	0.0252	0.0248	-0.0236	5.36	0.000	1.000	mean
∞	0.16	0	0.150	4.53	0.1023	0.0914	+0.0282	5.16	0.000	0.999	$\text{rev}_{0.05}$
∞	0.16	1	0.147	4.54	0.0985	0.0892	-0.0511	5.37	0.000	0.999	$\text{rev}_{0.2}$
∞	0.41	0	0.244	4.52	0.2684	0.2049	+0.0716	5.11	0.000	0.996	mean
∞	0.41	1	0.240	4.53	0.2606	0.1996	+0.0386	5.22	0.000	0.995	$\text{rev}_{0.05}$

S11. CHANNEL GEOMETRY

A. Design and calibration

The noise enters through two physically distinct channels: a static detuning δ of standard deviation σ_δ and a multiplicative drive-amplitude error of standard deviation σ_a, independent. We parameterize them by a mixing angle, $\sigma_\delta = s\cos\theta$ and $\sigma_a = s\sin\theta/\Omega_0$ with Ω_0 the nominal Rabi rate, and step θ through nine values from 0° to 90° in 11.25° increments. At each θ the overall scale s is recalibrated by a secant search so that the mean loss of the mean-optimal control matches a common target; without that step the two channels would differ in both geometry and strength and the comparison would be confounded. Each probe of the search is a single short optimization (120

iterations) rather than a full multistart, with one full multistart run at the calibrated scale. Two seeds, $T_g = 12.7$ ns, two-level model: 18 points. The achieved mean losses are 3.1–8.8 $\times 10^{-4}$ for $\theta < 90°$; at $\theta = 90°$ the control nulls the amplitude response almost entirely and the calibration cannot hold, giving 5–6 $\times 10^{-6}$.

B. Pre-registered predictions and outcomes

B1: The measured tail-to-mean ratio follows $\rho_\alpha(m_{\text{eff}})$ as the channel split moves m_{eff}, with Spearman > 0.8 and median relative error below 10%. *Outcome: fails as written, passes when blocked on seed.* Pooled Spearman = $+0.712$ ($p = 9 \times 10^{-4}$), median error 7.2%. Within seeds, $+0.976$ ($p = 3 \times 10^{-5}$) and $+0.952$ ($p = 3 \times 10^{-4}$) at median errors 7.9% and 3.4%; seed-centred over the 16 in-regime points, $+0.962$ ($p = 3 \times 10^{-9}$). The pooled statistic is defeated by a constant offset of $+0.563 \pm 0.141$ between the seeds against a prediction that varies by only 1.51 across the whole range of m_{eff}. The rank correlation of m_{eff} with the measured ratio is -0.976 and -0.952 against a predicted -1 exactly.

B2: Where the split moves m_{eff} by more than 0.3, the CDFs of the mean- and CVaR-optimal controls cross. *Outcome: not scored.* The design produced one such pair, so the test has no power. We report this rather than reporting "0 of 1 crossings".

B3: The κ^2 envelope continues to hold, $\max R_{\text{tail}} < 0.10$. *Outcome: fails on* 18 *points, holds on the* 16 *in regime.* The two failures are the $\theta = 90°$ pair, at $\kappa = 6.26$ and 46.85: a bound of the form $C\kappa^2 + O(\kappa^3)$ is not a statement about $\kappa = 47$. Excluding them, all remaining points have $\kappa \leq 0.116$ and $\max R_{\text{tail}} = 0.0275$.

B4: One curve $\rho_\alpha(m_{\text{eff}})$ fits both this route and the colored-noise route of Sec. S10; the two residual sets are not distinguishable. *Outcome: passes.* Mann–Whitney $p = 0.77$ on the 16 in-regime points against the 18 Stage 4 points with $R^2_{\text{fit}} \geq 0.94$ ($p = 0.40$ including the $\theta = 90°$ pair). Medians -1.5% and -0.1%; 10th–90th percentiles $[-8.0, +3.9]\%$ and $[-7.4, +2.6]\%$. Pooled over all 34 points, Spearman = $+0.831$ ($p = 1 \times 10^{-9}$) at a median absolute residual of 3.7%.

C. Per-point table

θ in degrees; σ_a is the relative amplitude error; ratio and prediction at $\alpha = 0.05$; R_{tail} against the best mean comparator. The last two rows are the excluded $\theta = 90°$ pair.

θ	seed	σ_δ	σ_a	m_{eff}	κ	ratio	pred.	R_{tail}
0.00	0	0.0196	0.0000	1.0000	0.078	5.141	5.582	+0.0089
0.00	1	0.0217	0.0000	1.0000	0.094	5.760	5.582	+0.0275
11.25	0	0.0187	0.0009	1.0021	0.088	5.164	5.576	-0.0120
11.25	1	0.0208	0.0011	1.0013	0.084	5.781	5.578	+0.0253
22.50	0	0.0191	0.0020	1.0060	0.078	5.114	5.565	+0.0058
22.50	1	0.0197	0.0021	1.0094	0.094	5.732	5.556	+0.0147
33.75	0	0.0240	0.0041	1.0245	0.116	5.079	5.515	+0.0103
33.75	1	0.0196	0.0033	1.0146	0.080	5.756	5.542	+0.0173
45.00	0	0.0194	0.0049	1.0354	0.082	5.048	5.486	+0.0055
45.00	1	0.0177	0.0045	1.0343	0.075	5.710	5.489	-0.0748
56.25	0	0.0185	0.0070	1.0725	0.078	4.959	5.391	+0.0024
56.25	1	0.0190	0.0072	1.1240	0.097	5.489	5.268	+0.0111
67.50	0	0.0182	0.0112	1.2110	0.089	4.701	5.078	-0.0009
67.50	1	0.0236	0.0144	1.1953	0.113	5.275	5.110	+0.0139
78.75	0	0.0178	0.0228	1.7566	0.111	3.990	4.245	-0.1975
78.75	1	0.0166	0.0213	1.9205	0.116	4.197	4.072	-0.0616
90.00	0	0.0000	0.0469	1.0000	46.850	12.908	5.582	+0.3655
90.00	1	0.0000	0.0478	1.0000	6.256	11.640	5.582	+0.4175

S12. TRAJECTORY-LEVEL RELAXATION

A. An exact one-jump decomposition

Amplitude damping is not a Gaussian perturbation of the propagator, so the machinery of Sec. S1 does not apply and the loss distribution must be constructed directly. Monte Carlo wavefunction sampling would do it, but the quantity we need — a tail average at $\alpha = 10^{-4}$ — is precisely the one trajectory sampling estimates worst. We instead expand in the number of jumps and keep every branch exactly.

Let a be the lowering operator, $n = a^\dagger a$ and $\Gamma = 1/T_1$. On each control slice we use the Strang-split damped map

$$K_k = D^{1/2} U_k D^{1/2}, \qquad D = \exp(-\tfrac{1}{2}\Gamma\, dt\, n), \tag{S28}$$

so that the no-jump branch is $K_N \cdots K_1$. A single jump at slice k sends the state to the normalized collapsed state, after which the evolution is again no-jump. The observable for that branch is $\mathrm{Tr}(O\, S_k \sigma S_k^\dagger) = \mathrm{Tr}\big((S_k^\dagger O S_k)\sigma\big)$ with $S_k = K_N \cdots K_{k+1}$, so all N one-jump branches are obtained from a single backward recursion that accumulates $S_k^\dagger O S_k$. The cost is one extra pass over the time grid, and the truncation error is $O((\Gamma T)^2)$.

Two details decide whether this agrees with the Lindblad channel, and both were caught by validation rather than by inspection.

Bayes weighting. The four input states of the average-gate-fidelity estimator do not have the same probability of producing a given record: the no-jump weight is $\mathrm{Tr}(K\rho_j K^\dagger)$, which depends on j. A plain average over inputs is therefore inconsistent with the channel and leaves an $O(\Gamma)$ discrepancy. Weighting each input by $P(j \mid r) \propto \mathrm{Tr}(K_r \rho_j K_r^\dagger)$ removes it: the residual disagreement with a direct Lindblad solve falls from 2.3×10^{-5} to 1.2×10^{-6} at $T_1 = 30\,\mu$s, which is 0.46% of the T_1 contribution itself.

The jump weight. The natural weight $\Gamma\, dt\, \langle n \rangle$ is inconsistent with the *discretized* no-jump map at $O(\Gamma\, dt)$, which shows up as a drifting mass defect. The consistent choice is the telescoping trace loss of the map actually used, $m_{jk} = \mathrm{Tr}\,\rho_j(t_k) - \mathrm{Tr}\,K_k \rho_j(t_k) K_k^\dagger$, placed on the normalized collapsed state. With it, total probability is conserved to $0.8\,(\Gamma T)^2$ with no floor in dt. Across the whole study the largest residual mass defect is 2.2×10^{-6}.

Validation. Six self-tests run before every production job: exactness of the atomic CVaR against a sorted brute-force reference including the boundary-atom split (1.6×10^{-15}); the Γ-scaling of the discrepancy against a Lindblad reference; mass conservation against $(\Gamma T)^2$; agreement of ρ_α and Ξ_α with their closed forms; the mixture identity of Eq. (8) on a two-channel toy; and the backward recursion against a direct $O(N^2)$ construction (6.7×10^{-14}). The last of these caught an off-by-one in the recursion — the update $S_k^\dagger O S_k$ must be applied *before* storing, since $S_k = S_{k+1} U_k$ — which had produced a maximum relative error of 39.

B. Design and scoring

The grid crosses $T_g \in \{12.7, 40\}$ ns, $T_1 \in \{10, 30\}\,\mu$s and $\sigma_\phi^2 \in \{0.04, 0.16\}$ with two seeds on the two-level model, with $\sigma_\phi^2 = 0.16$ run at $T_g = 40$ ns only, plus a four-level arm at $T_g = 12.7$ ns, $T_1 = 10\,\mu$s, $\sigma_\phi^2 = 0.04$: 14 points. Tail objectives are optimized at $\alpha \in \{0.20, 0.05, 0.01\}$ and every control is evaluated at $\alpha \in \{0.5, 0.2, 0.05, 0.01, 2 \times 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$. The fairness protocol of Sec. S9.A is unchanged: common warm start, budget parity, reverse warm starts, and R_{tail} scored against the best mean-side comparator. The comparator upgraded 13 of the 14 cells, by at most 0.188 — a larger fraction than any earlier stage, which is itself a statement about how flat the landscape is here.

C. Pre-registered predictions and outcomes

P1: The shift in the integrated excited population between the mean and tail optima scales as $\Xi_\alpha = 1/(\alpha\rho_\alpha)$. *Outcome: the estimator is mis-specified; we do not score the prediction.* The through-origin slope of $(n_{\mathrm{free}} - n_{\mathrm{CVaR}})$ on $(n_{\mathrm{free}} - n_{\mathrm{mean}})$ is 0.716 (bootstrap CI $[0.254, 3.790]$) against a predicted 3.583, and it does not move as the resolution gate is tightened: 0.716 at eleven points, 0.709 at eight, 0.660 at five. Regression dilution would lift it; it does not. The identity presumed that n_{int} is the coordinate along which the dephasing/jump trade-off is resolved, and the spread table below shows it is not. The sign agrees at 13 of 14 points.

P2: β orders R_{tail} and κ becomes inert. *Outcome: the second half holds, the first fails.* Spearman$(\beta, R_{\mathrm{tail}}) = +0.222$ ($p = 0.45$), and the same in every subset: $+0.112$ on the twelve two-level points, $+0.168$ dropping the two points

with $\beta > 10$, +0.011 scoring against the raw mean comparator. Spearman$(\kappa, R_{\text{tail}}) = +0.249$ ($p = 0.39$); partial given β, +0.178 ($p = 0.56$). As in Sec. S10, the replicate scatter caps any such correlation: run-to-run standard deviation of R_{tail} is 0.0316 against a total spread of 0.0402, an attenuation ceiling of 0.62, so the disattenuated values are +0.36 and +0.40. Neither is significant.

P3: Exceedances of the κ^2 envelope are expected, since Stage 5 breaks (A1) by construction; not scored as a failure. *Outcome:* $\max R_{\text{tail}} = 0.030$ with six of 14 positive. No large exceedance occurred.

P4: The tail-to-mean ratio follows Eq. (8). *Outcome: passes.* Median relative error 1.2% over the twelve two-level points, every point above r_α, across β from 0.098 to 63. We record explicitly that the reported Spearman(pred, meas) = +0.991 is *not* an independent check — the prediction is a monotone function of β alone, so it equals Spearman(β, ratio) to machine precision. The content of the test is the level agreement at no free parameters. The four-level arm degrades to 9.9% and 16.7%, in both cases with the measurement below the prediction, as a third (leakage) channel would require.

D. Weight and leverage

The reason P2 fails is visible in how little of the jump channel the control can reach. Over the eight controls optimized at each point — free, mean, three tail levels, three reverse warm starts — the fractional spread (max − min)/mean is a median 0.9% for the jump loss ℓ_j (worst case 2.0%), a median 1.4% for n_{int} (worst 3.5%), and a median 45% for the dephasing loss they are traded against (worst 573%). The jump probability is pinned by the gate's own requirement on the integrated excited population, and the cost of a relaxation event is set by the target unitary. A tail objective multiplies the weight on that channel by $\Xi_\alpha = 3.58$; there is nothing for the extra weight to act on.

Two cells at $\sigma_\phi^2 = 0.16$ make the point sharply. At seed 1 the optimizer reaches a basin in which the linear phase response is almost completely cancelled: $\mathbb{E}L_{\text{deph}}$ falls by a factor of 73 against the seed-0 sibling, the mean loss improves from 2.82×10^{-3} to 1.35×10^{-3}, κ rises from 0.128 to 6.21 because $\|A\|_F$ collapses, and β reaches 63. All eight controls at that point land in the same basin, so it is a property of the problem and not of one restart. R_{tail} there is −0.0008.

E. Per-point table

T_g in ns, T_1 in μs, ratio at $\alpha = 0.05$, R_{tail} against the best mean comparator.

T_g	T_1	σ_ϕ^2	lv	seed	$10^3 p$	ℓ_j	β	κ	ratio	R_{tail}
12.7	10	0.04	2	0	0.61	0.676	1.924	0.070	14.92	+0.0058
12.7	10	0.04	2	1	0.61	0.675	1.809	0.062	14.95	+0.0036
12.7	30	0.04	2	0	0.20	0.677	0.648	0.070	10.99	+0.0020
12.7	30	0.04	2	1	0.20	0.674	0.359	0.062	9.54	-0.1285
40.0	10	0.04	2	0	1.88	0.667	2.893	0.060	16.16	-0.0413
40.0	10	0.04	2	1	1.90	0.671	3.345	0.060	16.71	+0.0120
40.0	10	0.16	2	0	1.91	0.671	0.836	0.128	11.83	-0.0300
40.0	10	0.16	2	1	1.99	0.669	63.072	6.215	19.87	-0.0008
40.0	30	0.04	2	0	0.64	0.673	1.280	0.063	13.48	-0.0121
40.0	30	0.04	2	1	0.64	0.672	0.947	0.060	12.65	-0.0675
40.0	30	0.16	2	0	0.63	0.667	0.263	0.121	8.20	+0.0302
40.0	30	0.16	2	1	0.67	0.670	21.709	6.024	19.65	+0.0009
12.7	10	0.04	4	0	0.56	0.577	0.099	0.036	5.74	-0.0014
12.7	10	0.04	4	1	0.56	0.576	0.098	0.036	6.19	-0.0140

S13. THE MEASUREMENT-ONLY PREDICTOR

This section gives the construction behind Sec. IV L of the main text, the protocol correction it forced, and the gating rules used to score it.

A. Construction

Let $\{\epsilon_k\}_{k=1}^{K}$ be the error coordinates of the test ensemble under a control u, with weights w_k, losses $L_k = \epsilon_k^{\mathsf{T}} W \epsilon_k$, and let $T_\alpha = \{k : L_k \geq \mathrm{VaR}_\alpha\}$ carry total weight α. The tail-conditional displacement is

$$d_\alpha = \frac{1}{\alpha} \sum_{k \in T_\alpha} w_k \epsilon_k - \sum_k w_k \epsilon_k, \tag{S-disp}$$

and the object of interest is the change in CVaR_α under a common offset $\epsilon_k \mapsto \epsilon_k - t\, d_\alpha$ applied to every draw.

Three corrections separate a usable predictor from the naive one.

(i) Curvature. The naive Hessian of CVaR_α in the offset is $2W$, which ignores that the membership of T_α changes as the offset moves. Writing m_c for the tail mass and Σ_τ for the covariance of the error coordinates in a narrow band around the VaR level, with ρ_τ the density of loss there, the correct curvature is

$$H = \frac{m_c}{\alpha}\, 2W + \frac{4\rho_\tau}{\alpha}\, W \Sigma_\tau W. \tag{S-hess}$$

The band is widened adaptively until it contains enough draws to estimate Σ_τ, and a flag is raised when it cannot. The boundary term is not a refinement: omitting it over-predicts the gain by 315% in a brute-force comparison on the synthetic reduced model, which is how the error was found.

(ii) Reachability. The offset must lie in $\mathcal{B} = \mathrm{range}(\partial\epsilon/\partial c)$, computed by finite differences on the control coefficients and whitened in the W metric. On the real simulator this projector has rank 2 of 3 at every point—one error coordinate is unreachable—though the measured displacement already lies in $\mathcal{B}$ to machine precision (reach frac = 1.000), so the rank deficiency is not the source of the calibration constant.

(iii) Step length. Rather than trusting the quadratic, the step is set by a line search of CVaR_α along d_α over a seven-decade two-sided grid with local refinement. Because the search and the evaluation would otherwise use the same draws, the search is cross-fitted. The fold split is mixture-aware: splitting naively puts a rare atom entirely in one fold, leaving that fold with no heavy tail and a spurious gain, so responsive draws are split while inert branches are kept in both folds with renormalized weights.

We report three nested statistics per point. $R_{\mathrm{pred}}^{\mathrm{simple}}$ uses $H = 2W$ and no line search; $R_{\mathrm{pred}}^{\mathrm{quad}}$ uses Eq. (S-hess); and R_{pred} adds the cross-fitted line search. Against a brute-force re-optimization on the synthetic model their relative errors are +315%, $\{16, 77, 47\}$% and $\{0.6, 3.3, 4.4\}$% at $\kappa = \{0.1, 0.25, 0.5\}$. Only the last is used for the calibration in the main text.

B. Resolution floor

A displacement estimated from αK draws has sampling scatter even when the true gain is zero, so every point carries a floor R_{min} built from the band statistics and the effective tail count, and a point is called resolved when $R_{\mathrm{pred}} > R_{\mathrm{min}}$. On 24 independent null draw sets the floor matches the empirical scatter to a ratio of 0.75. The floor drives the test-ensemble size: the working rule is $\alpha_{\mathrm{min}}\, n_{\mathrm{test}} \gtrsim 400$, which with $\alpha_{\mathrm{min}} = 0.025$ gives $n_{\mathrm{test}} = 50\,000$ and is enforced rather than advised. The training ensemble obeys a separate rule, $\alpha_{\mathrm{min}} K_{\mathrm{train}} \geq 8$, because the CVaR objective's gradient is built only from its own training tail; at the default $K_{\mathrm{train}} = 128$ that is 3.2 draws at $\alpha = 0.025$. One matched ensemble is shared by the mean, CVaR and reverse runs at a point, since giving the CVaR arm more data than its comparator would break the budget parity the fairness protocol rests on.

C. The shared basin pool

The comparator of Sec. IV A—best of the mean control and the reverse warm starts—is not budget-neutral when the mean multistart has not converged. Using the criterion that no other control at a point beats the mean control on its own objective by more than 5%, only 6 of 16 cells were converged, the worst at a ratio of 0.250. Choosing the comparator by lowest mean loss rather than lowest CVaR gives numerically identical results at every point, so this is not selection on the outcome variable; it is an unequal restart budget.

The pooled protocol collects every control any arm found at a point (nine, deduplicating to three or four), and runs each objective from each start, selecting each arm's winner on the validation ensemble by that arm's own risk measure. Table S16 shows the effect on the replicate scatter. Because every start is warm, the pooled pass costs roughly len(pool) × 5 single optimizations against the 36 the original point cost.

TABLE S16. Replicate scatter in the measured gain between the two seeds, before and after equalizing the restart budget. The pooled protocol removes almost all of it, which is why the slope in Table IV is a calibration rather than noise.

α	0.20	0.10	0.05	0.025
best-of-comparator	0.272	0.249	0.178	0.136
shared basin pool	0.010	0.016	0.033	0.019
reduction factor	26.8	15.2	5.4	7.1

D. Gating

Two gates are applied and both are reported alongside the ungated statistics.

For the direction test, a point enters if it is resolved, if the CVaR arm did not find a better mean optimum than the mean control (mean-loss ratio ≥ 0.95), and if $|R^{\mathrm{raw}}| > 0.005$. The first two conditions say that the CVaR control is a local displacement of the mean control rather than a different basin; the third says there is a displacement at all. This removes 4 to 6 of 16 points depending on α. The pass criterion is significant anti-alignment—no point with $\cos < 0$ whose null probability is below 0.05—rather than a minimum cosine, since in three dimensions cos is uniform on $[-1, 1]$ under the null and a single low value carries almost no information.

For the level test, no gating is applied: all 16 points enter at every α, and the same-basin and raw-comparator variants are reported beside the pooled one.

E. The two leverage arms

The relaxation arm is analytic. Stage 5 never sampled jumps; it used the exact no-jump/one-jump decomposition, recording $(p, \ell_j, \mathbb{E}L_c)$ and hence β per cell. The reconstruction appends one atom of weight p carrying loss ℓ_j, with its error coordinates pinned at ϵ_0 so that it contributes nothing to the displacement, and rescales ℓ_j so that the re-derived cell's β matches the recorded one—β, not the absolute jump loss, being the invariant Proposition 4 is written in. Two cells at $\kappa = 7.4$ are excluded from the identity test as far outside the perturbative regime; they satisfy it anyway. We note that cells sharing $(T_g, \sigma_\phi^2, \mathrm{seed}, \mathrm{levels})$ but differing in T_1 produce identical re-derived controls, since T_1 does not enter the dephasing bulk—so the 14 rows are 7 distinct bulk controls paired with two relaxation rates, and the effective count for any statement about the bulk is 7.

The fluctuator arm installs a two-state telegraph process in the simulator, so that the optimizer, the losses and the propagators all see it, and sweeps two axes: excursion amplitude at fixed rarity, and rarity at fixed amplitude. At each point the predictor is evaluated twice—once on the mixture and once on the same control restricted to the non-fluctuator draws—so the suppression ratio is taken at one control and one κ and is not confounded by the fact that amplitude drives κ and rarity in opposite directions. A rank correlation between Λ and R_{pred} across an amplitude sweep is -1 by construction and is reported only to make that visible.

F. Validation

Twelve checks run without the simulator and are required to pass before any point is computed: exact atomic CVaR at four α; the vanishing of the debiased displacement in the linear-Gaussian limit and the $1/K$ decay of the raw bias; the closed form against brute-force re-optimization at three κ; the predicted direction; the κ^2 onset; two leverage checks against brute force; the floor against empirical null scatter; the two closed-form identities for ρ_α; and the mixture identity of Eq. (20) against its analytic form, which agrees to 2.4% and reproduces the multiplicative relation to 1.0% on the synthetic model.

S14. PER-POINT RESULTS

A. All validated comparisons (E1–E8)

Every cell is re-evaluated in the full Lindblad model on the test ensemble. R_{tail} is against the best mean control; "same-budget" is against the multistart mean control alone; "upgraded" marks cells in which a reverse start improved

the comparator by more than 0.1%. T_g in ns, $\sigma_\delta/2\pi$ in MHz. Point labels: *Tg*, gate-time sweep; *2lv*, two levels; *sig*, phase-variance sweep.

Source	Point	T_g	$\sigma_\delta/2\pi$	α	R_{tail} vs best mean [95% CI]	same-budget	comparator
2lvl_long	Tg-2lv 00	20.8	2.50	0.05	−0.007 [-0.021, +0.009]	+0.098	upgraded
2lvl_long	Tg-2lv 00	20.8	2.50	0.2	−0.082 [-0.094, -0.071]	+0.026	upgraded
2lvl_long	Tg-2lv 01	32.5	2.50	0.05	+0.009 [-0.015, +0.033]	+0.009	multistart
2lvl_long	Tg-2lv 01	32.5	2.50	0.2	+0.009 [+0.003, +0.015]	+0.009	multistart
amp_only	Tg 00	12.7	0.00	0.05	−0.013 [-0.018, -0.007]	−0.007	upgraded
amp_only	Tg 00	12.7	0.00	0.2	−0.008 [-0.011, -0.006]	−0.004	upgraded
amp_only	Tg 01	15.4	0.00	0.05	−0.166 [-0.179, -0.154]	−0.007	upgraded
amp_only	Tg 01	15.4	0.00	0.2	−0.001 [-0.004, +0.001]	+0.112	upgraded
amp_only	Tg 02	20.8	0.00	0.05	−0.001 [-0.001, -0.001]	−0.002	multistart
amp_only	Tg 02	20.8	0.00	0.2	−0.001 [-0.001, -0.001]	−0.002	multistart
collapse_T20	sig-2lv 00	20.0	0.56	0.05	−0.001 [-0.003, +0.001]	+0.057	upgraded
collapse_T20	sig-2lv 00	20.0	0.56	0.2	−0.043 [-0.047, -0.040]	−0.000	upgraded
collapse_T20	sig-2lv 01	20.0	1.59	0.05	+0.003 [+0.002, +0.005]	+0.003	multistart
collapse_T20	sig-2lv 01	20.0	1.59	0.2	+0.002 [-0.004, +0.009]	+0.002	multistart
collapse_T20	sig-2lv 02	20.0	3.18	0.05	+0.021 [+0.018, +0.024]	+0.021	multistart
collapse_T20	sig-2lv 02	20.0	3.18	0.2	+0.010 [+0.002, +0.019]	+0.010	multistart
collapse_T20	sig-2lv 03	20.0	5.03	0.05	+0.056 [+0.049, +0.062]	+0.067	upgraded
collapse_T20	sig-2lv 03	20.0	5.03	0.2	+0.013 [+0.003, +0.024]	+0.026	upgraded
collapse_T20	sig-2lv 04	20.0	10.07	0.05	+0.292 [+0.247, +0.335]	+0.773	upgraded
collapse_T20	sig-2lv 04	20.0	10.07	0.2	−3.556 [-4.292, -3.023]	+0.105	upgraded
collapse_T8	sig-2lv 00	8.0	1.41	0.05	−0.002 [-0.005, +0.001]	+0.130	upgraded
collapse_T8	sig-2lv 00	8.0	1.41	0.2	−0.119 [-0.126, -0.111]	+0.000	upgraded
collapse_T8	sig-2lv 01	8.0	3.98	0.05	+0.001 [-0.004, +0.005]	+0.001	multistart
collapse_T8	sig-2lv 01	8.0	3.98	0.2	+0.002 [-0.006, +0.011]	+0.002	multistart
collapse_T8	sig-2lv 02	8.0	7.96	0.05	+0.018 [+0.013, +0.022]	+0.018	multistart
collapse_T8	sig-2lv 02	8.0	7.96	0.2	+0.010 [+0.001, +0.019]	+0.010	multistart
collapse_T8	sig-2lv 03	8.0	12.57	0.05	+0.051 [+0.045, +0.056]	+0.051	multistart
collapse_T8	sig-2lv 03	8.0	12.57	0.2	+0.024 [+0.014, +0.034]	+0.024	multistart
collapse_T8	sig-2lv 04	8.0	25.17	0.05	+0.187 [+0.142, +0.224]	+0.187	multistart
collapse_T8	sig-2lv 04	8.0	25.17	0.2	+0.095 [+0.079, +0.108]	+0.095	multistart
det_only	Tg 00	12.7	2.50	0.05	−0.015 [-0.035, +0.005]	−0.011	upgraded
det_only	Tg 00	12.7	2.50	0.2	−0.001 [-0.010, +0.009]	+0.003	upgraded
det_only	Tg 01	15.4	2.50	0.05	+0.074 [+0.058, +0.086]	+0.154	upgraded
det_only	Tg 01	15.4	2.50	0.2	+0.034 [+0.019, +0.047]	+0.111	upgraded
det_only	Tg 02	20.8	2.50	0.05	+0.177 [+0.153, +0.195]	+0.167	upgraded
det_only	Tg 02	20.8	2.50	0.2	+0.068 [+0.050, +0.085]	+0.077	upgraded
fast	audit	12.7	2.50	0.01	−0.047 [-0.160, +0.056]	−0.038	upgraded
fast	audit	12.7	2.50	0.025	−0.011 [-0.025, +0.003]	−0.000	upgraded
fast	audit	12.7	2.50	0.05	−0.020 [-0.043, -0.001]	−0.010	upgraded
fast	audit	12.7	2.50	0.1	−0.004 [-0.014, +0.007]	+0.006	upgraded
fast	audit	12.7	2.50	0.2	−0.013 [-0.025, -0.001]	−0.004	upgraded
fast	sig-2lv 00	12.7	0.50	0.05	−0.001 [-0.002, -0.001]	+0.038	upgraded
fast	sig-2lv 00	12.7	0.50	0.2	−0.026 [-0.028, -0.024]	−0.000	upgraded
fast	sig-2lv 01	12.7	1.00	0.05	−0.002 [-0.005, +0.001]	−0.002	multistart
fast	sig-2lv 01	12.7	1.00	0.2	+0.000 [-0.003, +0.003]	+0.000	multistart
fast	sig-2lv 02	12.7	2.50	0.05	+0.002 [-0.003, +0.007]	+0.002	multistart
fast	sig-2lv 02	12.7	2.50	0.2	+0.003 [-0.004, +0.009]	+0.002	multistart
fast	sig-2lv 03	12.7	5.00	0.05	+0.029 [+0.026, +0.032]	+0.029	multistart
fast	sig-2lv 03	12.7	5.00	0.2	+0.013 [+0.004, +0.022]	+0.013	multistart
fast	sig-2lv 04	12.7	8.00	0.05	+0.077 [+0.068, +0.085]	+0.077	multistart
fast	sig-2lv 04	12.7	8.00	0.2	+0.034 [+0.024, +0.045]	+0.034	multistart
fast	sig-2lv 05	12.7	16.00	0.05	+0.275 [+0.247, +0.298]	+0.275	multistart
fast	sig-2lv 05	12.7	16.00	0.2	+0.135 [+0.116, +0.152]	+0.135	multistart
fast	Tg 00	6.3	2.50	0.05	−0.003 [-0.007, +0.002]	−0.002	upgraded
fast	Tg 00	6.3	2.50	0.2	−0.002 [-0.014, +0.010]	+0.001	upgraded
fast	Tg 01	7.4	2.50	0.05	−0.011 [-0.024, +0.003]	−0.011	multistart
fast	Tg 01	7.4	2.50	0.2	+0.001 [-0.006, +0.007]	+0.001	multistart
fast	Tg 02	8.5	2.50	0.05	−0.016 [-0.030, -0.001]	−0.013	upgraded
fast	Tg 02	8.5	2.50	0.2	−0.002 [-0.011, +0.006]	−0.000	upgraded
fast	Tg 03	9.9	2.50	0.05	−0.017 [-0.031, -0.001]	−0.016	multistart
fast	Tg 03	9.9	2.50	0.2	−0.002 [-0.011, +0.008]	−0.001	multistart
fast	Tg 04	11.5	2.50	0.05	−0.021 [-0.041, -0.002]	−0.021	multistart
fast	Tg 04	11.5	2.50	0.2	−0.004 [-0.016, +0.008]	−0.004	multistart
fast	Tg 05	12.7	2.50	0.05	−0.022 [-0.046, -0.001]	−0.013	upgraded
fast	Tg 05	12.7	2.50	0.2	−0.013 [-0.026, -0.000]	−0.006	upgraded
fast	Tg 06	15.4	2.50	0.05	+0.104 [+0.080, +0.125]	+0.107	upgraded
fast	Tg 06	15.4	2.50	0.2	+0.038 [+0.027, +0.049]	+0.042	upgraded
fast	Tg 07	20.8	2.50	0.05	+0.102 [+0.074, +0.123]	+0.102	multistart

Source	Point	T_g	$\sigma_\delta/2\pi$	α	R_{tail} vs best mean [95% CI]	same-budget	comparator
fast	Tg 07	20.8	2.50	0.2	+0.069 [+0.057, +0.079]	+0.069	multistart
fast	Tg 08	32.5	2.50	0.05	+0.064 [+0.035, +0.090]	+0.079	upgraded
fast	Tg 08	32.5	2.50	0.2	−0.022 [-0.034, -0.009]	+0.024	upgraded
fast	Tg-2lv 00	6.3	2.50	0.05	−0.026 [-0.053, -0.000]	−0.026	multistart
fast	Tg-2lv 00	6.3	2.50	0.2	−0.000 [-0.003, +0.003]	−0.000	multistart
fast	Tg-2lv 01	12.7	2.50	0.05	−0.026 [-0.053, +0.003]	−0.026	multistart
fast	Tg-2lv 01	12.7	2.50	0.2	−0.005 [-0.011, +0.001]	−0.005	multistart
fast	Tg-2lv 02	15.4	2.50	0.05	−0.012 [-0.030, +0.004]	−0.012	multistart
fast	Tg-2lv 02	15.4	2.50	0.2	−0.000 [-0.003, +0.004]	−0.000	multistart
seed1	audit	12.7	2.50	0.01	−0.145 [-0.176, -0.080]	−0.137	upgraded
seed1	audit	12.7	2.50	0.025	−0.047 [-0.082, -0.010]	−0.041	upgraded
seed1	audit	12.7	2.50	0.05	−0.041 [-0.075, -0.011]	−0.034	upgraded
seed1	audit	12.7	2.50	0.1	−0.044 [-0.072, -0.017]	−0.038	upgraded
seed1	audit	12.7	2.50	0.2	−0.015 [-0.020, -0.011]	−0.010	upgraded
seed1	sig-2lv 00	12.7	5.00	0.05	−0.396 [-0.430, -0.362]	+0.050	upgraded
seed1	sig-2lv 00	12.7	5.00	0.2	+0.005 [+0.000, +0.010]	+0.318	upgraded
seed1	sig-2lv 01	12.7	16.00	0.05	+0.271 [+0.219, +0.316]	+0.271	multistart
seed1	sig-2lv 01	12.7	16.00	0.2	+0.097 [+0.059, +0.127]	+0.097	multistart
seed2	audit	12.7	2.50	0.01	+0.014 [+0.001, +0.029]	+0.016	upgraded
seed2	audit	12.7	2.50	0.025	+0.010 [-0.017, +0.033]	+0.013	upgraded
seed2	audit	12.7	2.50	0.05	+0.015 [+0.000, +0.029]	+0.018	upgraded
seed2	audit	12.7	2.50	0.1	+0.003 [-0.008, +0.012]	+0.005	upgraded
seed2	audit	12.7	2.50	0.2	−0.004 [-0.012, +0.003]	−0.001	upgraded
seed2	sig-2lv 00	12.7	5.00	0.05	+0.036 [+0.024, +0.046]	+0.036	multistart
seed2	sig-2lv 00	12.7	5.00	0.2	+0.011 [+0.008, +0.014]	+0.011	multistart
seed2	sig-2lv 01	12.7	16.00	0.05	+0.314 [+0.279, +0.341]	+0.314	multistart
seed2	sig-2lv 01	12.7	16.00	0.2	+0.142 [+0.127, +0.154]	+0.142	multistart
stage1	ch σ_γ=0.0318 ρ=+0.99 s0	12.7	0.62	0.05	+0.018 [+0.010, +0.026]	+0.018	multistart
stage1	ch σ_γ=0.0318 ρ=+0.99 s0	12.7	0.62	0.2	+0.006 [+0.004, +0.007]	+0.006	multistart
stage1	ch σ_γ=0.127 ρ=+0 s0	12.7	0.00	0.05	+0.084 [-0.037, +0.175]	+0.447	upgraded
stage1	ch σ_γ=0.127 ρ=+0 s0	12.7	0.00	0.2	−0.192 [-0.235, -0.153]	+0.238	upgraded
stage1	ch σ_γ=0.0635 ρ=+0.99 s0	12.7	1.25	0.05	+0.139 [+0.095, +0.183]	+0.139	multistart
stage1	ch σ_γ=0.0635 ρ=+0.99 s0	12.7	1.25	0.2	+0.055 [+0.039, +0.070]	+0.056	multistart
stage1	ch σ_γ=0.01 ρ=+0 s0	12.7	2.50	0.05	−0.134 [-0.159, -0.106]	+0.194	upgraded
stage1	ch σ_γ=0.01 ρ=+0 s0	12.7	2.50	0.2	−0.348 [-0.360, -0.335]	+0.003	upgraded
stage1	ch σ_γ=0.02 ρ=+0 s0	12.7	2.50	0.05	−0.029 [-0.059, -0.001]	−0.029	multistart
stage1	ch σ_γ=0.02 ρ=+0 s0	12.7	2.50	0.2	−0.005 [-0.012, +0.001]	−0.005	multistart
stage1	ch σ_γ=0.04 ρ=+0 s0	12.7	2.50	0.05	−0.013 [-0.030, +0.005]	−0.013	multistart
stage1	ch σ_γ=0.04 ρ=+0 s0	12.7	2.50	0.2	+0.016 [+0.003, +0.028]	+0.016	multistart
stage1	ch σ_γ=0.08 ρ=+0 s0	12.7	2.50	0.05	+0.044 [+0.008, +0.084]	+0.044	multistart
stage1	ch σ_γ=0.08 ρ=+0 s0	12.7	2.50	0.2	+0.029 [+0.018, +0.042]	+0.029	multistart
stage1	ch σ_γ=0.08 ρ=+0 s1	12.7	2.50	0.05	+0.028 [+0.008, +0.045]	+0.028	multistart
stage1	ch σ_γ=0.08 ρ=+0 s1	12.7	2.50	0.2	+0.021 [+0.010, +0.032]	+0.021	multistart
stage1	ch σ_γ=0.08 ρ=+0 s2	12.7	2.50	0.05	+0.003 [-0.018, +0.020]	+0.003	multistart
stage1	ch σ_γ=0.08 ρ=+0 s2	12.7	2.50	0.2	+0.008 [+0.006, +0.010]	+0.008	multistart
stage1	ch σ_γ=0.127 ρ=+0.5 s0	12.7	2.50	0.05	+0.013 [-0.010, +0.035]	+0.013	multistart
stage1	ch σ_γ=0.127 ρ=+0.5 s0	12.7	2.50	0.2	+0.018 [+0.003, +0.031]	+0.018	multistart
stage1	ch σ_γ=0.127 ρ=+0.99 s0	12.7	2.50	0.05	+0.289 [+0.240, +0.343]	+0.288	multistart
stage1	ch σ_γ=0.127 ρ=+0.99 s0	12.7	2.50	0.2	+0.153 [+0.124, +0.179]	+0.153	multistart
stage1	ch σ_γ=0.127 ρ=+0.99 s1	12.7	2.50	0.05	+0.073 [-0.081, +0.196]	+0.073	multistart
stage1	ch σ_γ=0.127 ρ=+0.99 s1	12.7	2.50	0.2	−0.054 [-0.146, +0.015]	−0.054	multistart
stage1	ch σ_γ=0.127 ρ=+0.99 s2	12.7	2.50	0.05	+0.334 [+0.293, +0.373]	+0.349	upgraded
stage1	ch σ_γ=0.127 ρ=+0.99 s2	12.7	2.50	0.2	+0.133 [+0.111, +0.154]	+0.150	upgraded
stage1	ch σ_γ=0.127 ρ=+0.9 s0	12.7	2.50	0.05	+0.077 [+0.065, +0.089]	+0.077	multistart
stage1	ch σ_γ=0.127 ρ=+0.9 s0	12.7	2.50	0.2	+0.074 [+0.056, +0.091]	+0.074	multistart
stage1	ch σ_γ=0.127 ρ=+0 s0	12.7	2.50	0.05	+0.017 [-0.009, +0.039]	+0.017	multistart
stage1	ch σ_γ=0.127 ρ=+0 s0	12.7	2.50	0.2	+0.032 [+0.017, +0.046]	+0.032	multistart
stage1	ch σ_γ=0.127 ρ=-0.5 s0	12.7	2.50	0.05	+0.114 [+0.050, +0.168]	+0.114	multistart
stage1	ch σ_γ=0.127 ρ=-0.5 s0	12.7	2.50	0.2	+0.004 [-0.026, +0.030]	+0.004	multistart
stage1	ch σ_γ=0.127 ρ=-0.99 s0	12.7	2.50	0.05	+0.226 [+0.133, +0.300]	+0.224	multistart
stage1	ch σ_γ=0.127 ρ=-0.99 s0	12.7	2.50	0.2	+0.130 [+0.105, +0.152]	+0.130	multistart
stage1	ch σ_γ=0.127 ρ=-0.9 s0	12.7	2.50	0.05	+0.167 [+0.079, +0.244]	+0.171	upgraded
stage1	ch σ_γ=0.127 ρ=-0.9 s0	12.7	2.50	0.2	+0.084 [+0.055, +0.108]	+0.088	upgraded
stage1	ch σ_γ=0.127 ρ=+0.99 s0	20.0	1.59	0.05	+0.222 [+0.188, +0.258]	+0.244	upgraded
stage1	ch σ_γ=0.127 ρ=+0.99 s0	20.0	1.59	0.2	+0.136 [+0.104, +0.163]	+0.161	upgraded
stage1	ch σ_γ=0.127 ρ=+0.99 s0	8.0	3.97	0.05	+0.369 [+0.341, +0.398]	+0.369	multistart
stage1	ch σ_γ=0.127 ρ=+0.99 s0	8.0	3.97	0.2	+0.251 [+0.228, +0.273]	+0.251	multistart
strong	Tg 00	15.4	2.50	0.05	+0.100 [+0.077, +0.122]	+0.100	multistart
strong	Tg 00	15.4	2.50	0.2	+0.030 [+0.020, +0.039]	+0.030	multistart
strong	Tg 01	20.8	2.50	0.05	+0.117 [+0.092, +0.138]	+0.124	upgraded
strong	Tg 01	20.8	2.50	0.2	+0.060 [+0.046, +0.072]	+0.066	upgraded
strong	Tg 02	32.5	2.50	0.05	+0.059 [+0.036, +0.080]	+0.059	multistart

Source	Point	T_g	$\sigma_\delta/2\pi$	α	R_{tail} vs best mean [95% CI]	same-budget	comparator
strong	Tg 02	32.5	2.50	0.2	+0.027 [+0.017, +0.037]	+0.027	multistart
strong_2lvl	Tg-2lv 00	15.4	2.50	0.05	−0.008 [-0.038, +0.022]	+0.076	upgraded
strong_2lvl	Tg-2lv 00	15.4	2.50	0.2	−0.052 [-0.069, -0.033]	+0.000	upgraded
strong_2lvl	Tg-2lv 01	20.8	2.50	0.05	−0.021 [-0.043, -0.001]	−0.021	multistart
strong_2lvl	Tg-2lv 01	20.8	2.50	0.2	+0.000 [-0.001, +0.002]	+0.000	multistart
strong_2lvl	Tg-2lv 02	32.5	2.50	0.05	−0.004 [-0.010, +0.001]	−0.000	upgraded
strong_2lvl	Tg-2lv 02	32.5	2.50	0.2	−0.000 [-0.007, +0.006]	+0.004	upgraded

B. Dominance test (E9)

Closed-system statistics over each candidate pool (fresh 4000-draw ensemble) and the full-Lindblad $R_{\mathrm{tail}}(0.05)$ of the standard comparison against the best mean control. κ: second-order ratio of the mean control. τ: Kendall τ between mean and $\mathrm{CVaR}_{0.05}$ rankings. Rev.: fraction of confirmed rank reversals. Dom.: fraction of mean-separated pairs crossing for $q \geq 0.5$. Ratio: median $\mathrm{CVaR}_{0.05}$/mean (prediction 5.58). R_{pool}: cross-fitted pool regret. °Comparator upgraded by a reverse start.

Lev.	T_g	seed	σ_ϕ^2	κ	τ	Rev.	Dom.	Ratio	R_{pool}	R_{tail} Lindblad
2	8	0	0.0016	0.01	0.98	0.000	0.000	5.60	+0.000	−0.001°
2	8	0	0.0064	0.03	0.95	0.000	0.000	5.59	+0.000	−0.014°
2	8	0	0.04	0.07	0.80	0.001	0.000	5.52	+0.003	+0.001°
2	8	0	0.16	0.13	0.62	0.095	0.054	5.37	+0.022	+0.017°
2	8	0	0.41	0.21	0.47	0.226	0.256	4.95	+0.053	+0.052°
2	8	0	1.6	0.43	0.62	0.165	0.269	9.81	+0.178	+0.191°
2	12.7	0	0.0016	0.01	0.94	0.000	0.000	5.60	+0.000	−0.001°
2	12.7	1	0.0016	0.01	0.90	0.000	0.000	5.59	+0.000	−0.040°
2	12.7	2	0.0016	0.01	0.97	0.000	0.000	5.49	+0.000	−0.032°
2	12.7	0	0.0064	0.03	0.95	0.000	0.000	5.59	+0.000	−0.002
2	12.7	1	0.0064	0.03	0.95	0.000	0.000	5.59	+0.001	−0.013
2	12.7	2	0.0064	0.03	0.95	0.000	0.000	5.49	+0.001	+0.001
2	12.7	0	0.04	0.08	0.83	0.001	0.000	5.51	+0.002	+0.003°
2	12.7	1	0.04	0.08	0.87	0.008	0.000	5.52	+0.003	−0.040°
2	12.7	2	0.04	0.08	0.86	0.005	0.000	5.46	−0.001	−0.001°
2	12.7	0	0.16	0.15	0.65	0.125	0.040	5.50	+0.032	+0.027°
2	12.7	1	0.16	0.15	0.68	0.123	0.050	5.37	+0.028	+0.001°
2	12.7	2	0.16	0.15	0.63	0.105	0.052	5.38	+0.034	+0.033
2	12.7	0	0.41	0.25	0.59	0.190	0.192	5.36	+0.079	+0.072
2	12.7	1	0.41	0.25	0.55	0.238	0.273	5.46	+0.072	+0.044°
2	12.7	2	0.41	0.25	0.57	0.198	0.156	5.29	+0.090	+0.089°
2	12.7	0	1.6	0.52	0.29	0.364	0.468	4.82	+0.309	+0.274°
2	12.7	1	1.6	0.50	0.34	0.315	0.361	4.85	+0.292	+0.276°
2	12.7	2	1.6	0.52	0.10	0.452	0.607	4.61	+0.309	+0.290
2	20	0	0.0016	0.01	0.98	0.000	0.000	5.60	+0.000	−0.011°
2	20	0	0.0064	0.03	0.95	0.000	0.000	5.59	−0.002	−0.001
2	20	0	0.04	0.07	0.83	0.002	0.000	5.58	+0.003	+0.001°
2	20	0	0.16	0.14	0.56	0.135	0.027	5.54	+0.028	+0.022°
2	20	0	0.41	0.23	0.58	0.167	0.177	5.43	+0.071	+0.062°
2	20	0	1.6	0.48	0.10	0.473	0.675	11.04	+0.342	+0.353°
4	12.7	0	0.0016	0.01	0.92	0.009	0.013	1.88	+0.001	−0.003°
4	12.7	0	0.0064	0.02	0.90	0.010	0.015	3.06	+0.002	−0.013°
4	12.7	0	0.04	0.04	0.85	0.018	0.007	4.73	+0.014	−0.008
4	12.7	0	0.16	0.08	0.71	0.068	0.055	5.17	+0.040	+0.027
4	12.7	0	0.41	0.12	0.57	0.175	0.229	5.04	+0.081	+0.068
4	12.7	0	1.6	0.26	0.55	0.165	0.232	4.15	+0.048	+0.042°
4	20.8	0	0.0016	0.01	0.96	0.002	0.000	1.75	+0.000	−0.002°
4	20.8	0	0.0064	0.02	0.95	0.001	0.000	2.59	+0.001	−0.000°
4	20.8	0	0.04	0.06	0.74	0.046	0.026	3.32	+0.029	+0.016°
4	20.8	0	0.16	0.12	0.49	0.220	0.266	4.67	+0.123	+0.104°
4	20.8	0	0.41	0.20	0.20	0.380	0.527	5.16	+0.278	+0.274
4	20.8	0	1.6	0.38	0.19	0.397	0.579	4.93	+0.231	+0.240

S15. REPRODUCIBILITY

A. Code

All code is plain Python 3 with NumPy and SciPy (plus Matplotlib for the figures). Every step writes machine-readable output (JSON, CSV and NPZ with the full control vectors and per-draw test losses), so every table and figure can be regenerated from saved data without re-running an optimization.

File	Role
`fade_realistic.py`	transmon model, thermal state, Butterworth filter matrix, Hann seed pulse, worker pool
`stage0.py`	per-draw full-Lindblad worker with exact (Van Loan) gradients
`stage0_v2.py`	configuration, noise ensembles (with correlation), CVaR weights, bounded pulse basis, optimizer, fairness protocol, paired bootstrap, Stage-0 steps
`fast_viper.py`	closed-system Daleckii–Krein simulator, continuation starts, self-test against Lindblad, full-Lindblad `validate` step with the best-mean comparator
`stage1.py`	Stage-1 `channels` runs and the `diagnose` step (error rotation vectors, loss-quadratic fits, error clouds)
`run_final.sh`	the final Stage-0 batch (collapse test, replication, strong search, validation)
`make_figures.py`	Stage-0 figures and the Sec. S14 table
`make_stage1_figures.py`	κ, reduced model, offsets, independence test
`make_paper_extras.py`	explanatory figures, pulses, loss distributions, error clouds, simulator check
`stage2.py`, `run_stage2.sh`	Stage-2 candidate pools and dominance statistics
`make_stage2_figures.py`	Stage-2 figures, revised dominance test, κ, table
`make_kappa_bound.py`	the κ^2 law: closed-form coefficients, the scalar and random-geometry checks, and the comparison with every diagnosed control
`stage3.py`, `run_stage3.sh`	the controlled κ experiment, leakage-aware diagnosis and the generality grid
`stage3_recomp.py`	applies the best-mean comparator to the saved closed-system losses
`make_stage3_figures.py`	Stage-3 figures and the partial-correlation analysis

Built-in checks. `stage0_v2.py selftest` and `fast_viper.py selftest` must pass before any production run. They check gradients against finite differences, the closed-system model against the Lindblad code with dissipation switched off (10^{-7}), that dissipation acts as a near-constant shift, and that the CVaR weights reproduce the exact estimator. `stage1.py` refuses to run with an older version of the ensemble code that lacks the correlation parameter.

B. Commands

Stage 0 (from the project folder, 8 worker processes):

```
export FADE_WORKERS=8
python fast_viper.py selftest                # default outdir results_fast
python fast_viper.py characterize            # leakage onset
python fast_viper.py audit --Tg-audit 12.7
python fast_viper.py sweep --Tg 6.337 7.351 8.529 9.894 11.479 12.7 \
       15.449 20.793 32.467                  # E2, 4 levels
python fast_viper.py sweep --levels 2 --Tg 6.337 12.7 15.449
python fast_viper.py sweep --levels 2 --Tg 20.793 32.467 \
       --outdir results_2lvl_long
python fast_viper.py sweep --Tg 12.7 15.449 20.793 --sigma-d-mhz 0 \
       --outdir results_amp_only             # E3
python fast_viper.py sweep --Tg 12.7 15.449 20.793 --sigma-a 0 \
       --outdir results_det_only
python fast_viper.py sigma --levels 2 --Tg-sigma 12.7 \
       --sigma-grid 0.5 1 2.5 5 8 16         # E4 at 12.7 ns
python fast_viper.py sweep --levels 2 --Tg 15.449 20.793 32.467 --restarts 20 \
       --maxiter 1000 --outdir results_strong_2lvl
bash run_final.sh 2>&1 | tee final_log.txt   # collapse, seeds, strong search
```

(`fast_viper.py` passes every step except `selftest` and `validate` to `stage0_v2.py` with the fast simulator patched in.) `run_final.sh` contains, among others,

```
python fast_viper.py sigma --levels 2 --Tg-sigma 8 \
       --sigma-grid 1.41 3.98 7.96 12.57 25.17 --outdir results_collapse_T8
python fast_viper.py sigma --levels 2 --Tg-sigma 20 \
```

```
       --sigma-grid 0.56 1.59 3.18 5.03 10.07 --outdir results_collapse_T20
python fast_viper.py audit --Tg-audit 12.7 --seed $s --outdir results_seed$s
python fast_viper.py sweep --Tg 15.449 20.793 32.467 --restarts 20 \
       --maxiter 1000 --outdir results_strong
python fast_viper.py validate --outdir <folder>     # for every folder
```

Stage 1 (2 levels, $T_g = 12.7$ ns unless stated):

```
python stage1.py channels --sigma-a-grid 0.01 0.02 0.04 0.08 0.127
python stage1.py channels --sigma-a-grid 0.127 \
       --rho-grid -0.99 -0.9 -0.5 0.5 0.9 0.99
python stage1.py channels --sigma-d-mhz 0 --sigma-a-grid 0.127
python stage1.py channels --sigma-a-grid 0.08 --seed 1     # and --seed 2
python stage1.py channels --sigma-a-grid 0.127 --rho-grid 0.99 --seed 1
python fast_viper.py validate --outdir results_stage1
python stage1.py diagnose results_stage1 results_fast ...
```

The prospective test (E8) used the same `channels` step at $\rho = 0.99$ with (noise $\times 0.25$) `--sigma-d-mhz 0.625 --sigma-a-grid 0.03175`, (noise $\times 0.5$) `--sigma-d-mhz 1.25 --sigma-a-grid 0.0635`, and at matched phase variance `--Tg 8 --sigma-d-mhz 3.97 --sigma-a-grid 0.127` and `--Tg 20 --sigma-d-mhz 1.5875 --sigma-a-grid 0.127`, each followed by `validate` and `diagnose`. The restart seeds of a `channels` point depend on its position in the grid of that call and on `--seed`; they are recorded in each point's JSON output.

Stage 2 (dominance test, overnight):

```
python stage2.py selftest                 # synthetic checks of every statistic
bash run_stage2.sh 2>&1 | tee stage2_log.txt
python stage2.py summary                  # per-point table
```

`run_stage2.sh` runs `stage2.py run` for the three grids of E9, then `validate`, `diagnose` and `summary`. The pool ensemble uses seed $44 + 1000s$.

Figures and tables:

```
python make_figures.py              # Stage 0 figures, per-point table
python make_stage1_figures.py       # kappa, model, offsets, independence
python make_paper_extras.py         # explanatory and Stage-1 detail figures
python make_stage2_figures.py       # Stage-2 figures, table, statistics
```

C. Seeds

Training, validation and test ensembles use seeds 11, 22 and 33, shifted by $1000s$ for replication seed s. Restart perturbations use a separate seed per objective and point. Bootstraps use 2000 resamples with seed 7 in the full-Lindblad validation. The reduced-model Monte Carlo uses 6×10^4 draws with seed 1, and the pre-compensation toy 2×10^6 draws with seed 0.

[1] R. T. Rockafellar and S. Uryasev, Optimization of conditional value-at-risk, J. Risk **2**(3), 21 (2000).
[2] C. Acerbi and D. Tasche, On the coherence of expected shortfall, J. Bank. Finance **26**, 1487 (2002).
[3] M. A. Nielsen, A simple formula for the average gate fidelity of a quantum dynamical operation, Phys. Lett. A **303**, 249 (2002).
[4] M. Shaked and J. G. Shanthikumar, *Stochastic Orders* (Springer, New York, 2007).
[5] A. W. Marshall, I. Olkin, and B. C. Arnold, *Inequalities: Theory of Majorization and Its Applications*, 2nd ed. (Springer, New York, 2011).
[6] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, Charge-insensitive qubit design derived from the Cooper pair box, Phys. Rev. A **76**, 042319 (2007).
[7] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, A quantum engineer's guide to superconducting qubits, Appl. Phys. Rev. **6**, 021318 (2019).
[8] F. Motzoi, J. M. Gambetta, P. Rebentrost, and F. K. Wilhelm, Simple pulses for elimination of leakage in weakly nonlinear qubits, Phys. Rev. Lett. **103**, 110501 (2009).
[9] C. F. Van Loan, Computing integrals involving the matrix exponential, IEEE Trans. Autom. Control **23**, 395 (1978).

[10] N. J. Higham, *Functions of Matrices: Theory and Computation* (SIAM, Philadelphia, 2008).
[11] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu, A limited memory algorithm for bound constrained optimization, SIAM J. Sci. Comput. **16**, 1190 (1995).
[12] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap* (Chapman & Hall, New York, 1993).